U(2)-Invariant Ricci Solitons on Four Manifolds

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Originality statement

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William Hadden
August, 2024

Abstract

In a series of papers published between 2002-2003 ([37], [38], [39]), Grisha Perelman was able to resolve the Poincaré conjecture by addressing finite time singularities in Hamilton's Ricci flow [23] on 3-dimensional Riemannian manifolds. This achievement raised the question of whether a similar topological classification exists for four-dimensional manifolds. This thesis explores a *class* of self-similar solutions to the Ricci flow, which are conjectured to model singularities in four dimensions. These self-similar solutions posses a large isometry group and thus the corresponding equations reduce to a system of ordinary differential equations characterised by a single parameter, α. Through the application of analytical and numerical techniques in conjunction with center manifold theory, we present strong evidence supporting A. Appleton's conjecture that at most one of these self-similar solutions can be a singularity model for finite time singularities. As a consequence, we present numerical algorithms to understand center manifolds more generally.

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Chapter 1

Introduction

1.1 Motivation

The Ricci flow is a geometric partial differential equation introduced by Hamilton in 1982 [23]. This non-linear, weakly parabolic flow evolves an initial Riemannian metric, g_0 by generating a sequence of time dependent metrics, $\{g_t\}_{t\in I}$ via

$$\frac{\partial g_t}{\partial t} = -2\mathrm{Ric}(g_t) \tag{1.1}$$

where $\mathrm{Ric}(g_t)$ is the Ricci curvature of g_t. This evolution equation was famously used by Perelman in the resolution of Thurston's geometrisation conjecture for 3-manifolds [37], [39] which solves the Poincaré conjecture as a special case [38].

The goal of the Ricci flow is to transform an initial metric, g_0, into a metric, g_T, that is distinguished by its curvature. Intuitively, one can think of the Ricci flow as analogous to the heat equation, in the sense that it spreads the Ricci *curvature* out in space. For instance, R. Hamilton in [23] showed that compact three-manifolds with positive Ricci curvature are such that the (unique) maximal solution of (1.1) once suitably re-normalised, converges to a metric of constant sectional curvature. The *Killing-Hopf* theorem ([28]-[24]) then asserts that such manifolds are diffeomorphic to a quotient S^3/Γ called a *space form* where Γ is a finite group of isometries such that the action of Γ on M has trivial stabiliser groups. Hence, any compact three-manifold, M with positive Ricci curvature is diffeomorphic to a space form. This example shows that while the Ricci flow is a *geometric* quantity, it allows one to prove *topological* statements in certain cases.

In three dimensions, the Ricci flow appears to "seek out" constant curvature metrics. In higher dimensions however, there are many more candidates for solutions to the Ricci flow. For instance, metrics of constant Ricci curvature called *Einstein metrics* given by

$$\mathrm{Ric}(g_t) = \lambda g_t \tag{1.2}$$

for some $\lambda \in \mathbb{R}$ and Riemannian metric, g_t where the underlying manifold is at least three dimensional. Indeed, if we suppose that our initial metric, g_0 is Einstein, then a solution $\{g_t\}_{i\in I}$ of (1.1) is given by

$$g_t = (1 - 2\lambda t)g_0. \tag{1.3}$$

For instance, the round (unit) sphere, S^n equipped with the round metric, g_0 satisfies $\mathrm{Ric}(g_0) = (n-1)g_0$. Thus, (1.3) implies that $g_t = (1 - 2(n-1)t)g_0$ is a solution to (1.1) and hence the sphere shrinks homothetically to the point at time $T = \frac{1}{2(n-1)}$ [44].

Einstein metrics can be naturally generalised to solutions of (1.1) that evolve only by scalings and diffeomorphisms called *Ricci Solitons*. Explicitly, these are solutions to (1.1) that satisfy,

$$g_t = \sigma(t)\varphi_t^*(g_0) \tag{1.4}$$

where $\sigma(t)$ is a positive, smooth function and $\varphi_t : M \to M$ is a 1-parameter family of diffeomorphisms. One can then show that the initial metric, g_0 of a Ricci soliton solution, g_t satisfies

$$\mathrm{Ric}(g_0) + \frac{1}{2}\mathcal{L}_X(g_0) = \lambda g_0 \tag{1.5}$$

where X is a vector field on M and $\lambda \in \mathbb{R}$. Notice we recover the Einstein condition (1.2) if X is the zero vector field, which we term *trivial* solutions. Recall that the round sphere, S^n equipped with the round metric, g_0 satisfied $\lambda > 0$. Moreover, S^n shrank homothetically under the Ricci flow to a point. This motivates classifying Ricci solitons as shrinking, steady or expanding if λ is positive, zero or negative respectively. Pictorially,

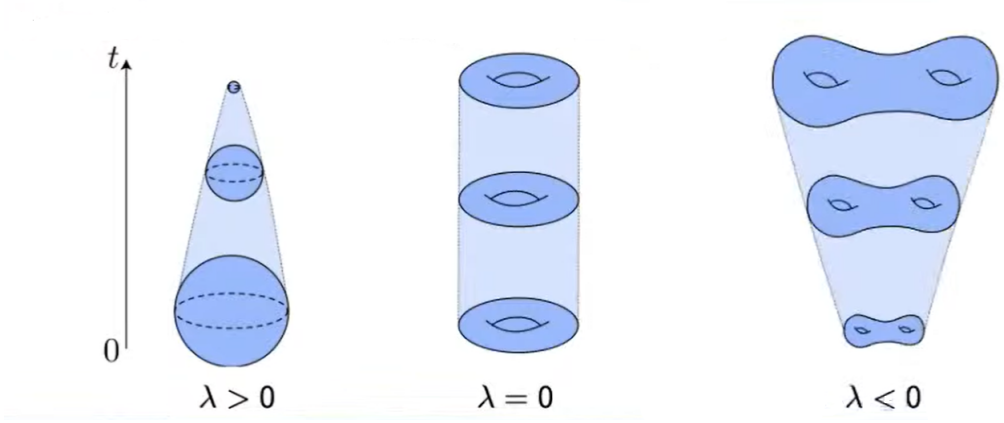


Figure 1.1: 3-dimensional Ricci solitons for different values of λ [2].

Further classes of solutions to (1.5) can be found by considering the vector field, X in (1.5). For instance, if X is the gradient of some scalar potential function, $u : M \to \mathbb{R}$, then the equality $\mathcal{L}_X(g) = 2\mathrm{Hess}_g(u)$ implies that (1.5) becomes

$$\mathrm{Ric}(g) + \mathrm{Hess}(u) = \lambda g \tag{1.6}$$

where $\mathrm{Hess}(u)$ denotes the Hessian of u. A solution (g, u) of (1.6) is then called a "Ricci Soliton of gradient type" or *gradient soliton*.

In general, (1.6) is a system of partial differential equations. However, one can use *symmetry assumptions* to obtain a system of ordinary differential equations, dependent on some initial

conditions (i.e. an initial value problem). These symmetry assumptions are interpreted in terms of *metric preserving* maps called *isometries*. It turns out that the collection of all these transformations can be associated to a Lie group [35]. For instance, if g is $U(2)-$invariant, then $h^*g = g$ where $h \in U(2)$ and h^*g denotes the pullback of g by h.

To directly relate symmetry assumptions to (1.6), suppose g_0 has a compact subgroup, G of isometries. Then, the potential function, $u : M \to \mathbb{R}$ is also invariant under the action of G in the sense that $u(p) = u(h \cdot p)$, $h \in G$ [41, Section 2]. Intuitively, one might expect $\mathrm{Hess}(u)$ to simplify and thereby simplify the analysis of (1.6).

Ricci solitons are strongly linked to the analysis of curvature blow-ups, called *singularities* in the Ricci flow. Heuristically, if $T < \infty$ is a singular time, then we "zoom-in" on the solution in spacetime by re-scaling the metric, g_T according to the curvature. By zooming-in arbitrarily close, one often finds that the resulting object is a Ricci soliton. In other words, we can model singularities in the Ricci flow via Ricci solitons. For instance, Figure (1.2) shows a 3-sphere encountering a variety of singularities where the "zoomed-in" objects, termed *blow up limits*, are shown in green.

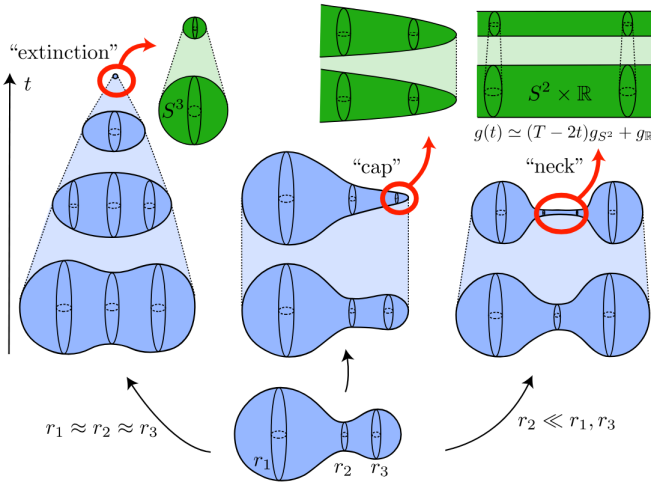


Figure 1.2: Singularity formation in the Ricci flow on a 3-sphere [2].

With regard to Figure 1.2, notice that the left-most blow-up limit resembles a sphere shrinking homothetically to the point. Heuristically, one might expect this limit to be modelled on the *shrinking* Ricci soliton we found earlier. Moreover, the middle blow-up limit can be identified as a rotationally symmetric, *steady* gradient soliton called the "Bryant soliton" as studied in [8] which opens asymptotically like a paraboloid. It turns out that different solitons model different *types* of singularities and that not every soliton is a singularity model.

For instance, suppose that (M, g_t) is a complete Ricci flow with maximal existence time, $T < \infty$. Then, a *Type 1 singularity* occurs at T if there exists some constant $C > 0$ such that

$$\sup_{p \in M} |\mathrm{Rm}(g_t)| \leq \frac{C}{T-t}$$

for all $t < T$. Otherwise, the singularity is of *Type-2*. It turns out that Type-1 singularities correspond to shrinking solitons while all *currently* known Type-2 singularities are modelled on steady solitons. For instance, the authors of [21] showed that the Bryant soliton is the unique, rotationally symmetric, steady Ricci soliton on $\mathbb{R}^n$ for $n \geq 3$ which encounters a finite time singularity.

Perelman introduced the concept of κ-non-collapsing in [37] to study singularity formation. Intuitively, a soliton is non-collapsed if it "does not lose a dimension" when rescaled according to curvature. We make this precise by considering the volume of balls on the soliton. Indeed, a Riemannian manifold, (M^n, g) is $\kappa-$non-collapsed on all scales, if for any point $p \in M$, and scale $r > 0$, we have

$$\frac{\mathrm{Vol}(B(p,r))}{r^n} \geq \kappa$$

whenever $|Rm| \leq \frac{1}{r^2}$ for all $y \in B(p,r)$ where $\kappa > 0$ and $|Rm|$ is the Frobenius norm extended to the Riemannian curvature tensor. We call a soliton that satisfies this hypothesis *non-collapsed*. Perelman showed that *only* non-collapsed solitons can model finite time singularities of compact Ricci flows ([37, No Collapsing Theorem]). Under certain curvature assumptions ([37, Section 11]), the same is true for non-compact solitons. For instance, Figure 1.2 illustrates that the Bryant soliton is a *candidate* for a singularity model and indeed we show that this soliton is non-collapsed in Section (3.5)[1]. On the other hand, the cigar soliton is collapsed and therefore cannot appear as such a singularity model (see [18] for further details).

In three dimensions, singularity models have been completely classified ([5], [6]) into three basic types. Explicitly, the shrinking round space forms, S^3/Γ the shrinking round cylinder, $S^2 \times \mathbb{R}$ and the Bryant soliton. In the four-dimensional case, the number of potential singularity models is much larger and such a classification has not been obtained. Indeed, constructing examples of singularity models is an active area of research. For instance, Feldman, Ilmanen and Knopf in [19] found a shrinking soliton called the *FIK shrinker*. Prior to this, H. Cao in [12] found a $U(2)-$invariant steady Kähler-Ricci soliton. However, it was shown to be collapsed and thus by [37], cannot appear as a blow-up limit. Recently, A. Appleton in [1] found a $U(2)-$invariant, non-collapsed, steady, gradient soliton which we term *Appleton's soliton* by considering a certain $U(2)$ group action on manifolds of dimension at least four. As a result, the soliton equations (1.6) reduce to a system of singular ordinary differential equations, characterised by a single parameter, α. Indeed, the solutions depend continuously on α which is linked to the geometry of the manifold at the singular orbit. Specifically, α uniquely determines the Hessian of the potential function, u at the singular orbit (Section 3). By smoothly varying α, Appleton was able to show that there exists a particular choice of α such that the solution is non-collapsed. Based on this observation, Appleton made the following conjecture in [1].

Conjecture 1.1.1. *For all $k \geq 3$ there exists a unique $\alpha^* \in \mathbb{R}$ such that the corresponding solitons are:*

1. *incomplete for all $\alpha > \alpha^*$,*

[1] Appleton's soliton has the same asymptotic properties as the Bryant soliton [1].

2. *complete and non-collapsed when* $\alpha = \alpha^*$,

3. *complete and collapsed when* $\alpha < \alpha^*$.

We investigate this conjecture and present strong numerical evidence for its truth. Furthermore, we study how one can construct and analyse the steady, gradient solitons that arise from the constructions proposed in [1] and [43].

1.2 Results and technical discussion

We consider the homogeneous vector bundle, M of the form

$$M = SU(2) \times_{U(1)} \mathbb{R}^2 \tag{1.7}$$

derived from the group diagram $\mathbb{Z}_k \subset U(1) \subset SU(2)$ where the induced Riemannian submersions $\pi : SU(2)/\mathbb{Z}_k \to SU(2)/U(1)$ are the generalised Hopf fibrations for $k \in \mathbb{N}\backslash\{1, 2, 4\}$ as in Section 2.3.2. We note that this is an example of the *two summands case*, first studied by M. Buzano in [10], where the isotropy representation of the principal orbit, $SU(2)/\mathbb{Z}_k$ splits into two in-equivalent $Ad_{\mathbb{Z}_k}$-invariant irreducible real summands. Hence, the metric on the non-singular or *principal* component, M_0 is of the form

$$g_{M_0} = dr^2 + f_1(r)^2 g_S + f_2(r)^2 g_V \tag{1.8}$$

where $S = U(1)/\mathbb{Z}_k$ is the collapsing sphere and $V = SU(2)/U(1)$ the singular orbit. The resulting Ricci soliton equations for steady solitons are then of the form,

$$\frac{d}{dr}\frac{f_1'}{f_1} = -(-u' + tr(L))\frac{f_1'}{f_1} + \frac{f_1^2}{2f_2^2} \tag{1.9}$$

$$\frac{d}{dr}\frac{f_2'}{f_2} = -(-u' + tr(L))\frac{f_2'}{f_2} + \frac{1}{f_2^2} - \frac{f_1^2}{2f_2^4}. \tag{1.10}$$

In order to a construct a complete metric, we need to establish the long term existence of solutions. Indeed, a maximal solution to (1.6) with positive definite shape operator, L induces a complete metric [47]. Hence, obtaining control of the components, $\frac{f_i'}{f_i}$ of the shape operator is key to obtaining complete metrics, g. With this in mind, (1.9) implies that positivity of f_1'/f_1 is preserved along r. However, the same is not necessarily true for f_2'/f_2. Instead, notice that if the estimate

$$\frac{f_1}{f_2} < \sqrt{2}$$

holds, then positivity of f_2'/f_2 is preserved and therefore a complete metric is induced. The quantity $Q := \frac{f_1}{f_2}$ will form a cornerstone of our analysis and is perhaps the most natural scale-invariant quantity in this setting. Indeed, A. Appleton was able to show that Q determines the "non-collapsedness" of solutions to the soliton equations (1.9)-(1.10). In particular, A. Appleton in [1] showed that:

- if $\lim_{r\to\infty} Q = 1$, then the solution is complete and *non-collapsed*,
- if $\lim_{r\to\infty} Q = 0$ then the solution is complete and *collapsed*,
- if $Q > 1$ for any $r > 0$, then the solution is *incomplete*

and hence was able to show that there exists an α such that the solution is non-collapsed.

In order to extend (1.8) *smoothly* over the singular orbit at $r = 0$, one must impose certain smoothness conditions given by

$$\begin{gathered} f_1(0) = 0,\ f_1'(0) = 1,\ \text{and}\ f_2(0) = \overline{f_2} > 0,\ f_2'(0) = 0 \\ u(0) = 0,\ u'(0) = 0,\ u''(0) = C \leq 0 \end{gathered} \tag{1.11}$$

as shown in Section 3.2. Notice that these are *singular* initial conditions for the equations (1.9)-(1.10). For instance, the quantity f_1'/f_1 becomes unbounded at $r = 0$ and thus one must specify $u''(0)$ in order to get a unique solution. Indeed, one can show that the solutions to the soliton equations (1.9)-(1.10) are characterised by some parameter, $\gamma \in \mathbb{R}$ that depends on $u''(0)$ as explained in Section 3.3. In any case, the short time existence of such solutions to this initial value problem exist based on the work of M. Buzano [17] as shown in Section 3.3.

To analyse the long term behaviour of the solutions to the equations (1.9)-(1.10), T. Buttsworth examined the change of variables,

$$X_1 = \frac{1}{-u' + tr(L)} \frac{f_1'}{f_1},\ Y_1 = \frac{1}{-u' + tr(L)} \frac{f_1}{f_2^2} \tag{1.12}$$

$$X_2 = \frac{1}{-u' + tr(L)} \frac{f_2'}{f_2},\ Y_2 = \frac{1}{-u' + tr(L)} \frac{1}{f_1} \tag{1.13}$$

on the time re-parametrisation defined by $\frac{d}{ds} = \frac{1}{-u'+tr(L)} \frac{d}{dr}$. We claim that this is a natural change of variables. Indeed, variations of these variables have been studied in [47], [41] and [42]. The key observation of T. Buttsworth in [43], is that the origin in $\mathbb{R}^4$ is a critical point of the soliton equations in the variables (1.12)-(1.13) *and* that Appleton's non-collapsed soliton is convergent to this point. Indeed, if $U_0 \subset \mathbb{R}$ denotes the set of parameters, γ such that the corresponding solution to the soliton equations (1.9)-(1.10) is non-collapsed, then there exists open neighbourhoods $U \subset \mathbb{R}$ of U_0 that are convergent to some small neighbourhood about the origin [43]. As such, one is able to establish an equivalence between solutions in U and those that lie on a submanifold, $\mathcal{M}$ described in Section 4. The key point is that the long-term behaviour of the dynamics on $\mathcal{M}$ are particularly simple. Indeed, with respect to solutions to (1.9)-(1.10) one can make the following classification.

Proposition. *The solutions to the soliton equations (1.9)-(1.10) in $\mathcal{M}$ with respect to the variables (1.12) - (1.13) that satisfy the smoothness conditions (3.2) are such that:*

- *if $Y_1 > Y_2$, the solution is incomplete and blows up in finite time,*

- *if* $Y_1 < Y_2$*, then the solution is complete and collapsed,*
- *if* $Y_1 = Y_2$*, then the solution is complete and* non-collapsed.

As a result, one can use the equivalence of trajectories on $\mathcal{M}$ with solutions corresponding to $\gamma \in U$ in conjunction with (1.2) to determine the long term behaviour. However, this equivalence relies on a non-trivial mapping that is difficult to analyse analytically. Instead, we examine how this equivalence can be understood numerically as described in Chapter 4. This numerical understanding then allows us to provide strong evidence for Conjecture (1.1.1).

Chapter 2

Background

In this chapter, we introduce fundamental concepts in Riemannian geometry and Lie theory. We then apply these ideas to spaces with large isometry groups known as *homogeneous spaces*. We also outline previous work on the study of Ricci solitons in homogeneous spaces. Finally, we examine spaces foliated by homogeneous spaces of co-dimension one, referred to as *cohomogeneity one* manifolds. We assume that the reader has some background in elementary differential geometry such as charts, atlases, metrics, differential forms and the implicit function theorem.

2.1 Riemannian Geometry

In this section we give an overview of the necessary concepts from Riemannian geometry. We draw from and recommend the works of J. M. Lee in [30], and W. Merry in [33] for further reading.

2.1.1 A brief introduction to Riemannian geometry

Riemannian geometry begins with the study of *smooth manifolds*. These are paracompact, second-countable and Hausdorff topological spaces that locally resemble Euclidean space. The key idea is that for every point p on the smooth manifold M, one can always define an inner product, $\langle \cdot, \cdot \rangle_p$ on the *tangent space* at p denoted T_pM that varies *smoothly* across M. We call this smoothly varying function a *Riemannian metric*, denoted by g. As such, we consider Riemannian manifolds as the pair, (M, g).

The key usefulness of Riemannian metrics is that it allows the measurement of distances and angles, similar to the corresponding notions in Euclidean space. Recall that in Euclidean space, the length, L of a smooth path, $\gamma : I \to \mathbb{R}^n$ is found by integrating the velocity vector via,

$$L(\gamma) = \int_0^1 |\gamma'(t)| dt.$$

In Riemannian geometry, we get the same definition by considering the norm $|\cdot| : T_pM \to \mathbb{R}$ given by $|\cdot| := \sqrt{g(\cdot, \cdot)}$. Explicitly,

$$L(\gamma) = \int_0^1 \sqrt{g_{\gamma(t)}(\gamma'(t), \gamma'(t))}dt \tag{2.1}$$

and hence, a (connected) Riemannian manifold, (M, g) inherits the structure of a metric space. Indeed, the distance, d between two points, $x, y \in M$ is given by,

$$d(x, y) := \inf\{L(\gamma) : \gamma \text{ is a piece-wise smooth curve from } \gamma(0) = x, \gamma(1) = b\} \tag{2.2}$$

with length as in (2.1). The key point is that (M, d) is a metric space [30] and thus we say that a (connected) Riemannian manifold, (M, g) is complete if it is complete as a metric space, (M, d). For instance,

Example 2.1.1. Consider $\mathbb{R}^n$ equipped with the usual inner product on each tangent space $T_p\mathbb{R}^n$ under the canonical identification $T_p\mathbb{R}^n = \mathbb{R}^n$. Then in *standard coordinates*, the metric, g is of the form

$$g = \sum_i dx^i dx^i = \sum_i (dx^i)^2 = \delta_{ij} dx^i dx^j$$

and thus the matrix of g is given by $g_{ij} = \delta_{ij}$. Furthermore, $\mathbb{R}^n$ is a complete space as the distance function induced by g is simply the Euclidean distance.

For our purposes, a more useful notion of completeness is that of *geodesically complete* manifolds. We will precisely define geodesic curves later on. Intuitively, these are curves with zero *acceleration* that satisfy some second order equation system of ODEs called the *geodesic equation*. Hence, we can consider *maximal* geodesics in the sense of maximal intervals of existence to solutions of ODEs. With this terminology, we say that a (connected) manifold (M, g) is *geodesically complete* if any *maximal* geodesic $\gamma : I \to M$ is such that $I = (-\infty, \infty)$. Then by the Hopf-Rinow theorem, this is equivalent to completeness as a metric space. Non-examples are relatively easy to construct. For example, equip any proper open subset of $\mathbb{R}^n$ with the Euclidean metric. Then there are geodesics that reach the boundary in finite time.

A natural way of creating more examples of Riemannian manifolds is to take direct products of Riemannian manifolds.

Example 2.1.2. Suppose $(M_1, g_1), (M_2, g_2)$ are Riemannian manifolds. Then the direct product $M_1 \times M_2$ is equipped with the Riemannian metric, $g = g_1 \oplus g_2$, called the *product metric*, defined by

$$g(X_1 + X_2, Y_1 + Y_2) = g_1(X_1, Y_1) + g_2(X_2, Y_2)$$

where $X_i, Y_i \in T_{p_i}M_i$ under the canonical identification $T_{(p_1,p_2)}(M_1 \times M_2) = T_{p_1}M \oplus T_{p_2}M_2$. Consider the local coordinates, $(x^1, \ldots, x^n)$ for M_1 and $(x^{n+1}, \ldots, x^{n+m})$ for M_2. Then the metric $g = g_{ij}dx^i dx^j$ in the coordinates $(x^1, \ldots, x^{n+m})$ is a block diagonal matrix given by,

$$(g_{ij}) = \begin{pmatrix} (g_1)_{ij} & 0 \\ 0 & (g_2)_{ij} \end{pmatrix}.$$

where the matrices g_1, g_2 are is block diagonal in their respective coordinates.

One of the most fundamental aspect in geometry is *curvature*. However, curvature as defined via second derivatives in elementary differential geometry is insufficient in the setting of Riemannian manifolds.

To see why, consider a curve $\gamma : I \to M$. Suppose we wanted to define $\ddot{\gamma}(t_0)$ as the derivative of $\dot{\gamma}$ with respect to t at t_0. Then, one would have to consider the difference quotient of two vectors, $\dot{\gamma}(t)$ and $\dot{\gamma}(t_0)$. However, these live in distinct tangent spaces, $T_{\gamma(t)}M$ and $T_{\gamma(t_0)}M$ respectively. As such, it doesn't make sense to subtract them. Instead, $\dot{\gamma}(t)$ is an example of a *vector field along a curve*. That is, a smooth assignment of points on the curve $\gamma(t)$ to their corresponding velocity vectors, $\dot{\gamma}(t)$. One then defines the notion of a *connection* to "connect" different tangent spaces and thus allow the calculation of directional derivatives of vector fields.

Before we define the notion of a connection precisely, we require the following concepts. A *tangent bundle*, denoted TM is the result of "gluing" all the tangent spaces together. That is,

$$\begin{aligned} TM &= \sqcup_{p\in M} T_pM \\ &= \cup_{p\in M} \{p\} \times T_pM \\ &= \cup_{p\in M} \{(p,v) \mid v \in T_pM\} \\ &= \{(p,v) \mid p \in M\, v \in T_pM\}. \end{aligned}$$

Remark 2.1.1. *TM inherits a natural smooth structure from M. Indeed, if M^n is n-dimensional, then TM is $2n$-dimensional.*

Then, a *vector field* is a smooth map that associates to each point, $p \in M$ a vector, $v \in T_pM$. To make this notion precise, we define the notion of a *fibre bundle* as follows.

Definition 2.1.2. Let E, M and F be smooth manifolds and suppose that $\pi : E \to M$ is a smooth surjective map. Then, $\pi : E \to M$ is a fibre bundle over M with fibre, F if at every $x \in M$, there is a neighbourhood, U of x and a smooth map

$$\alpha : \pi^{-1}(U) \to F$$

such that

$$(\pi, \alpha) : \pi^{-1}(U) \to U \times F$$

is a diffeomorphism. In particular, we call α a *bundle chart*, E the *total space* of the bundle, M the *base space* and F the *fibre*.

Then we can define a *section* as

Definition 2.1.3. Let $\pi : E \to M$ be a fibre bundle. A *section* of E is a smooth map $s : M \to E$ such that $\pi \circ s = id$. That is, a smooth map $s : M \to E$ such that

$$s(x) \in E_x, \quad \forall x \in M$$

Then, the set of all sections is denoted $\Gamma(E)$.

Thus, we can identify vector fields, X on M as sections of the tangent bundle, TM, denoted $X \in \Gamma(TM)$ by considering the natural projection $\mathrm{Pr}_1 : TM \to M$. With these concepts, we can resolve what it means to be a *connection*.

Definition 2.1.4. A connection, ∇ on a Riemannian manifold is a mapping,

$$\nabla : \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM)$$
$$(X, Y) \mapsto \nabla_X Y$$

that satisfies:

1. $\nabla_X(Y + Z) = \nabla_X Y + \nabla_X Z$,
2. $\nabla_{h_1 X + h_2 Y} Z = h_1 \nabla_X Z + h_2 \nabla_Y Z$,
3. $\nabla_X(h_1 Y) = h_1 \nabla_X Y + X(h_1) Y$,

where $X, Y, Z \in \Gamma(TM)$, $h_1, h_2 \in C^\infty(M)$ and $X(h) := X(p)(h)$ for $p \in M$.

Indeed, $\nabla_X Y$ is the derivative of Y along X. Furthermore, $\nabla_{X_p} Y$ is referred to as the *directional derivative* of Y at p in the direction of X_p (where $X_p := X(p)$). It turns out that by imposing two further conditions on the connection, we get a unique choice of connection for Riemann manifolds called the *Levi-Civita* connection.

Theorem 2.1.5. *For a Riemannian manifold (M, g), there is a unique connection called the Levi-Civita connection that satisfies:*

1. $[X, Y] - (\nabla_X Y - \nabla_Y X) = 0$,
2. $X \cdot \langle Y, Z \rangle_g = \langle \nabla_X Y, Z \rangle_g + \langle X, \nabla_Y Z \rangle_g$.

These conditions are called the "Torsion-free" and "compatibility with g" conditions respectively.

The proof of this result involves demonstrating that a connection which satisfies both of the conditions in (2.1.5) is uniquely determined by the so-called *Koszul Formula*, given by

$$2\langle \nabla_X Y, Z \rangle = X\langle Y, Z \rangle + Y\langle X, Z \rangle - Z\langle X, Y \rangle + \langle [X, Y], Z \rangle - \langle [X, Z], Y \rangle - \langle [Y, Z], X \rangle. \quad (2.3)$$

Notice that (2.3) indicates that the Levi-Civita connection can be written explicitly in terms of the Riemannian metric by considering some basis of vector fields. Moreover, we can use (2.3) to compute the *Hessian* of a scalar potential function, $u : M \to \mathbb{R}$ defined as follow.

Definition 2.1.6. Let (M, g) be a Riemannian manifold and $u \in C^\infty(M)$ be a smooth function. The Hessian of u, denoted $Hess(u)$ is a map, $Hess(u) : \Gamma(TM) \times \Gamma(TM) \to C^\infty(M)$ defined by

$$Hess(u) : (X, Y) \mapsto g(\nabla_X \nabla u, Y)$$

where ∇u is defined as the vector field such that $g(Z, \nabla u) = Z(u)$ for all $Z \in \Gamma(TM)$ and is called the *gradient vector field* of u.

Remark 2.1.7. *The Hessian in Definition (2.1.6) is a generalisation of the familiar Hessian on $\mathbb{R}^n$. Indeed, the Hessian with respect to the Euclidean metric (2.1.1) satisfies $Hess(u)(\partial_{x^i}, \partial_{x^j}) = \partial_{x^i}\partial_{x^j}u$.*

From now on, we will work with only the Levi-Civita connection. With regards to our earlier discussion, we can define a geodesic as follows.

Definition 2.1.8. Let M be a smooth manifold and $I \subseteq \mathbb{R}$ an open interval. Then a smooth curve $\gamma : I \to M$ is called *geodesic* if $\nabla_{\gamma'}\gamma' = 0$.

2.1.2 Curvature

With our main geometrical tools introduced, we can now define the notion of curvature. In this section, we will assume the reader is familiar with tensors. Recall that tensors simply generalise the concepts of scalars, vectors and matrices from linear algebra. For instance, a $(0,0)$-tensor is a scalar and a $(1,0)$-tensor is a vector. With this terminology, we can define the notion of *Riemannian curvature* as follows.

Definition 2.1.9. The Riemannian curvature tensor R associated to a Riemann manifold (M, g) is a map

$$\mathrm{R} : \Gamma(TM) \times \Gamma(TM) \times \Gamma(TM) \to \Gamma(TM)$$

such that

$$\mathrm{R}(X, Y)Z := (\nabla_{[X,Y]} - (\nabla_X\nabla_Y - \nabla_Y\nabla_X))Z$$

we can make R into a (0,4)-tensor according to

$$\mathrm{Rm}(X, Y, Z, T) := \langle \mathrm{R}(X, Y)Z, T)\rangle_g$$

where $\langle\cdot,\cdot\rangle_g$ is the metric defined by g. In addition given two linearly independent tangent vectors $u, v \in T_pM$, we define the sectional curvature, K as

$$\mathrm{K}(u, v) := \frac{\mathrm{Rm}(u, v, u, v)}{\langle u, u\rangle_g\langle v, v\rangle_g - \langle u, v\rangle_g^2}.$$

Note that if $u, v \in T_pM$ are orthonormal, then the expression simplifies to

$$\mathrm{K}(u, v) = \mathrm{Rm}(u, v, u, v)$$

Remark 2.1.10. *We note that the relevant computations concerning R as a $(3,1)$-tensor field and as a $(0,4)$-tensor can be found in [30, Section, 7].*

Informally, one arrives at this definition by considering the *parallel transport* of vectors around loops. For instance, if a vector in Euclidean space is transported around a loop, it will return to the same orientation after it has been transported. However, on a curved surface, it may be deflected slightly after traversal. It turns out that parallel transport along geodesics in curved Riemannian spaces works in the same fashion as in Euclidean space, allowing one to make a true comparison *without* appealing to an embedding. The Riemannian curvature therefore, measures the failure of this process. We refer the interested reader to [30] for further details.

The Riemannian curvature tensor has inherent symmetries that are often helpful for computational purposes. Explicitly,

Proposition 2.1.11. *The Riemannian curvature tensor $Rm(X, Y, Z, T)$ has the following properties,*

- $R(X, Y, Z, T) = -R(Y, X, Z, T) = -R(X, Y, T, Z)$,
- $R(X, Y, Z, T) = R(Z, T, X, Y)$,
- $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$,

for $X, Y, Z, T \in \Gamma(TM)$.

Finally, we define how the Riemannian curvature behaves over product manifolds.

Proposition 2.1.12. *Let (M_1, g_1), (M_2, g_2) be Riemannian manifolds and $X \in \Gamma(TM)$ such that $X = X_1 + X_2$ via the projections $X_i \in \Gamma(TM_i)$. Then the Riemannian curvature tensor, Rm on the direct product $M_1 \times M_2$ as in (2.1.2) is given by*

$$Rm(X_1 + X_2, Y_1 + Y_2, Z_1 + Z_2, W_1 + W_2) = Rm_1(X_1, Y_1, Z_1, W_1) + Rm_2(X_2, Y_2, Z_2, W_2)$$

for all $X_i, Y_i, Z_i, W_i \in \Gamma(TM_i)$ where Rm_i denote the Riemannian curvature tensors on the components (M_i, g_i) for $i \in \{1, 2\}$.

2.1.3 Ricci and scalar curvature

The Ricci curvature is required for the computation of solitons (1.5) and can be defined via the Ricci curvature tensor as follows.

Definition 2.1.13. The Ricci curvature tensor, Ric is given by

$$\mathrm{Ric}(X, Y) = tr(\mathrm{Rm}(X, \cdot, Y, \cdot) \tag{2.4}$$

for $X, Y \in \Gamma(TM)$. Hence, if $\{E_i\}_{i \in I}$ is an orthonormal basis of T_pM, the expression (2.4) becomes

$$\mathrm{Ric}(X, Y) = \sum_{i \in I} \mathrm{Rm}(X, E_i, Y, E_i).$$

We can also think of the Ricci curvature component-wise as a $(0, 2)$-tensor given by

$$\mathrm{Ric} = R_{ij} dx^i \wedge dx^j$$

where

$$\mathbb{R}_{ij} = \sum_{j=1}^{n} R^k_{kij}$$

Remark 2.1.14. *The Ricci curvature inherits the symmetry condition,*

$$R(X,Y) = R(Y,X)$$

for $X, Y \in T_pM$ as a result of the symmetries of the Riemannian curvature tensor (2.1.11).

We also note that the Ricci curvature of a vector is equal to the sum of the sectional curvatures through it. Precisely,

Lemma 2.1.15. *If $X \in T_pM$ such that $\{X, E_2, \ldots, E_n\}$ is an orthonormal basis of T_pM, then*

$$Ric(X,X) = \sum_{i=2}^{n} K(X, E_i)$$

Proof. This follows from the definition. That is,

$$\mathrm{Ric}(X,X) = \mathrm{Rm}(X,X,X,X) + \sum_{i=2}^{n} \mathrm{Rm}(X,E_i,X,E_i) = \sum_{i=2}^{n} \mathrm{K}(X,E_i).$$

□

Heuristically, $\mathrm{Ric}(X,X)$ measures the deviation in volume between Euclidean space and the curved space in some small sector in the direction X. Indeed, if the volume of small geodesic ball is smaller (larger) than the Euclidean volume, then the Ricci curvature is positive (negative). As such, one might expect the Ricci curvature to control the local growth of geodesic balls in a manifold.

To define volume over Riemannian manifolds (M^n, g), we integrate over regions as in Euclidean geometry. However, as Riemannian spaces have curvature, the corresponding differential form must relate to the metric, g. We call this differential n-form the *volume form*. Before we define the volume form, recall that if $p \in M$, then we have a chart defined on a neighbourhood of p given by $\sigma_i : U_i \to O_i$ where the set of all U_i form an open cover of M and O_i are open sets in $\mathbb{R}^n$. Then in *local coordinates*, given by $x^i := \mathrm{Pr}_i \circ \sigma$, we define the volume form as follows.

Definition 2.1.16. In local coordinates, the volume form, dV of a Riemannian manifold is given by

$$dV = \sqrt{det(g_{ij})} dx^1 \wedge \cdots \wedge dx^n.$$

Then, the volume of a some patch, $U \subset M$ is given by

$$\int_U 1 dV.$$

We stated earlier that the Ricci curvature controls the "local growth of balls". With the definition of volume (2.1.16), we can make this precise. Recall that *geodesic normal coordinates* are local coordinates such that geodesics through p correspond to straight lines through the origin. Then [31, Theorem 1.7] states that,

Theorem 2.1.17. *Let $x = (x^1, x^2, \dots, x^n)$ be a point in geodesic normal coordinates in some neighbourhood of $p \in M$. Then, the coefficients of the Taylor expansion of $g_{ij}(x)$ about $x = 0$ can be expressed as*

$$g_{ij}(x) = \delta_{ij} + \frac{1}{3} R_{ijkl}(p) x^k x^l + \mathcal{O}(|x|^3)$$

Moreover, the volume form in Definition 2.1.16, is locally given by

$$dV = \left(1 + \frac{1}{6} R_{jk} x^j x^k + \mathcal{O}(|x|^3)\right) dx^1 \wedge \cdots \wedge dx^n$$

which shows that the volume growth depends (locally) on the Ricci curvature. The last notion of curvature we introduce is *scalar curvature*, denoted R. Note that this is distinguishable from the Riemannian curvature by the number of subscripts.

Definition 2.1.18. Let $\{E_i\}_{i=1}^n$ be an orthonormal basis for T_pM where $p \in M$. Then the scalar curvature, R is given by

$$R = \sum_{n=1}^{n} \mathrm{Ric}(E_i, E_i)$$

In particular, $R = tr(\mathrm{Ric})$ as matrices.

Intuitively, the scalar curvature represents the "average" curvature. Furthermore, there is a useful connection to the norm of the Riemannian curvature tensor, $|Rm| := \sqrt{R_{ijkl} R^{ijkl}}$.

Lemma 2.1.19. *Let R be the scalar curvature of a Riemannian manifold, (M, g) and $|Rm|$ the norm of the Riemannian curvature tensor, given by the usual ℓ^2 norm extended to tensors. Then*

$$R \leq n^2 |Rm|$$

where n is the dimension of the manifold M.

Proof. Recall that for $v \in \mathbb{R}^d$, comparison of p-norms implies $\|v\|_1 \leq \sqrt{d}\|v\|_2$. By considering an orthonormal frame on M^n, we can identify the components of the Riemann tensor R_{ijkl} as elements of $\mathbb{R}^{n^4}$ to obtain

$$R = \sum_{i,j} R_{ijij} \leq \sum_{i,j} |R_{ijij}| \leq \sum_{i,j,k,l} |R_{ijkl}| \leq n^2 \sqrt{\sum_{i,j,k,l} |R_{ijkl}|^2} = n^2 |Rm|.$$

□

While this is not an optimal bound, we note that in the context of κ-collapsed manifolds (2.1.20), the condition $|Rm| \leq \frac{1}{r^2}$ implies that $\frac{1}{n^2} R \leq |Rm| \leq \frac{1}{r^2}$. As R is generally easier to work with, this is often helpful in the analysis of collapsed solitons.

2.1.4 Collapsed manifolds

Determining whether or not a manifold is κ-collapsed is essential for understanding singularity formation in Ricci flow. Indeed, recall the following notion of κ-collapsed, from Chapter 1.

Definition 2.1.20. A Riemannian manifold, (M^n, g) is $\kappa-$non-collapsed on all scales, if for any point $p \in M$, and scale $r > 0$, we have

$$\frac{\text{Vol}(B(p,r))}{r^n} \geq \kappa$$

whenever $|\text{Rm}| \leq \frac{1}{r^2}$ for all $y \in B(p,r)$ where $\kappa > 0$. We call a soliton that satisfies this hypothesis *non-collapsed*.

Remark 2.1.21. *If the manifold is flat (i.e. vanishing curvature), this definition intuitively states that the volume of ball on the manifold must be at least as large as a ball in Euclidean space up to scalings.*

We give some examples of manifolds that are collapsed and non-collapsed to help build up an intuition. We start by considering flat surfaces as follows.

Example 2.1.3. Consider $\mathbb{R}^n$ equipped with the Euclidean metric g. Then, one can check that $|\text{Rm}|$ vanishes identically. Hence the condition in (2.1.20) must be satisfied for all $r > 0$. Recall also that the volume of a ball, $B(x,r)$ in $\mathbb{R}^n$ is given by $\frac{\pi^{n/2}}{(n-1)!}r^n$. Thus,

$$\frac{\text{Vol}B(x,r)}{r^n} > 0$$

and hence Euclidean n-space is non-collapsed.

This shows that if a manifold "looks like" Euclidean space and is "unbounded" in some sense, the manifold is non-collapsed. To show why both hypotheses are necessary, consider the following example.

Example 2.1.4. Consider the flat torus, $\mathbb{R}^2/\mathbb{Z}_2$ defined by gluing the opposite sides of the Euclidean rectangle. Then, since $|Rm| \equiv 0$, it follows that the volume of the ball of radius r must have volume at least κr^2 for all $r > 0$. However, a ball on the torus cannot have a volume larger than the torus itself. Hence, for sufficiently large r, the ratio in Definition 2.1.20 goes to zero and thus the flat torus is collapsed.

Another example of an asymptotically flat manifold is Hamilton's cigar soliton [22].

Example 2.1.5. Let $M = \mathbb{R}^2$ and write $g = \rho^2(dx^2 + dy^2)$. If we write g in terms of geodesic distance from the origin, s and polar angle, θ, we find that

$$g = ds^2 + \tanh^2(s)d\theta^2$$

Thus as s goes to infinity, $\tanh(s)^2 \to 1$ and thus the solution resembles a open cylinder at infinity. However, one can check that a cylinder has zero Riemannian curvature (2.1.12). Moreover, the volume of a ball of radius r, on a cylinder (for sufficiently large s), grows like $\mathcal{O}(s)$. Therefore, the ratio in (2.1.20) goes to zero as $s \to \infty$ and hence the cigar soliton is κ-collapsed.

This example begins to demonstrate that non-collapsed solitons must be such that their curvatures decay sufficiently slowly whilst simultaneously "growing" at a sufficiently fast rate. Indeed, this is precisely what happens in the case of the Bryant soliton [8] in dimensions $n \geq 3$ which we show in Section (3.5)[1]

2.2 Homogeneous spaces

Homogeneous spaces are a special class of Riemannian manifolds characterised by a high degree of symmetry, defined via isometries. Specifically, these symmetries involve a Lie group acting by isometries on a manifold. The extensive symmetry of these manifolds make them particularly appealing to study, as it allows the geometric properties at one point to be extended to many points within in the manifold. This greatly simplifies calculations, such as those involving curvature. To begin studying these spaces, it is essential to understand what an isometry is.

2.2.1 Isometries

Isometries make precise the idea that two spaces "look the same" geometrically. This leads to the definition of an isometry as follows.

Definition 2.2.1. Suppose $(M, g), (\tilde{M}, \tilde{g})$ are Riemann manifolds. Then an isometry from (M, g) to $(\tilde{M}, \tilde{g})$ is a diffeomorphism, $\varphi : M \to \tilde{M}$ such that $\varphi^* \tilde{g} = g$.

Note that φ^* denotes the pullback of $\tilde{g}$ by φ. That is, for vector fields $u, v \in TpM$, $\varphi^* \tilde{g}(u, v) = \tilde{g}(\varphi_* u, \varphi_* v)$ where φ_* denotes the differential of φ. Thus, we have an equivalent interpretation of an isometry, φ as a smooth bijection such that each differential map, $d\varphi_p : T_pM \to T_{\varphi(p)}\tilde{M}$ is a linear isometry. That is, for $v, w \in T_pM$,

$$\langle v, w \rangle_g = \langle d\varphi(v), d\varphi(w) \rangle_{\tilde{g}}.$$

We have already remarked that a flat metric is one that has zero curvature. However, we have an equivalent definition of flatness in the language of isometries.

Definition 2.2.2. A metric, g is flat if it is locally isometric to the Euclidean metric. More precisely, each $p \in M$ has a neighbourhood, U that is isometric to an open set in $\mathbb{R}^n$ equipped with the Euclidean metric.

Now we can define symmetry precisely. To this end, consider the set of all isometries from (M, g) to itself. One can show that this forms a group under composition called the *isometry group*, denoted $\mathrm{Iso}(M, g)$. It turns out that $\mathrm{Iso}(M, g)$ is a *Lie group*[2] in certain situations. Indeed, the famous Myers-Steenrod theorem [35] below makes this precise.

Theorem 2.2.3. *Let (M, g) be a Riemannian manifold with isometry group, $Iso(M, g)$ such that M has finitely many connected components. Then $Iso(M, g)$ is a (finite dimensional) Lie group that acts smoothly on M.*

[1] The non-collapsed solitons in Section 3.5 have the same asymptotic properties as the Bryant soliton. Hence the computation presented therein works in both cases.

[2] We make this notion precise in Section 2.2.2.

Remark 2.2.4. *The topology here is the compact open topology. That is, uniform convergence on compact subsets.*

For instance, consider S^2 equipped with the round metric. Then, this metric is preserved under the action of the orthogonal group, $O(n)$ as a subgroup of $\mathrm{GL}_n(\mathbb{R})$ which is itself a Lie group.

2.2.2 Lie groups and Lie algebras

Lie groups are smooth manifolds that have a compatible group structure. As a result, they come with built in symmetry. For instance, the circle group

$$S^1 = \{e^{i\theta} \mid \theta \in [0, 2\pi)\}$$

with group action given by multiplication and the general linear group, $\mathrm{GL}_n(\mathbb{R})$ of $n \times n$ invertible real matrices with matrix multiplication. One *key* property of Lie groups is that the left (or right) multiplication map, $L_g : G \to G$, $h \mapsto gh$ is a diffeomorphism. As a result, we can smoothly transport points in G to the identity, $e \in G$. Thus, it suffices to examine the associated tangent space analog, called the *Lie algebra* at e, denoted $\mathfrak{g}$ for the purposes of equipping G with a metric. Indeed, there is a one-to-one correspondence between inner products on $\mathfrak{g}$ and metrics on G that make L_g an isometry called *left-invariant metrics*. In this section, we define the necessary properties of Lie groups and Lie algebras with an eye towards left-invariant metrics. This section is drawn from [33] and [30].

Remark 2.2.5. *We note that the concepts introduced in the section can be equivalently formulated in terms of right actions.*

Definition 2.2.6. A Lie group is a group, G that is also a smooth manifold such that the left multiplication map, : $L_g : G \to G, h \mapsto gh$ and the inverse map, $L_g^{-1} : G \to G, g \mapsto g^{-1}$ are smooth.

The identification of Lie groups as smooth manifolds suggests the existence of an associated tangent space. This leads us to the notion of a Lie algebra.

Definition 2.2.7. A Lie algebra is a vector space $\mathfrak{g}$, along with a bi-linear map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ such that for all $x, y, z \in \mathfrak{g}$,

- $[x, y] = -[y, x]$,
- $[[x, y], z] + [[y, z], x] + [[z, x], y] = 0$,

where the second condition is known as the Jacobi identity.

According to [30], the bracket of two smooth vector fields on smooth manifold, M is a Lie bracket on the space of smooth vector fields, $\Gamma(TM)$. It follows therefore that $\Gamma(TM)$ is a Lie algebra. We now define the concept of a left-invariant vector field.

Definition 2.2.8. Let G be a Lie group. A vector field is *left-invariant* if $L_{g*}(X) = X$ for all $g \in G$. Then, the space of all left-invariant vector fields on G is called the *corresponding Lie algebra of* G, denoted $\mathfrak{g}$.

Note that one can show $[\cdot,\cdot]$ is also in $\mathfrak{g}$ so $\mathfrak{g}$ is indeed a Lie algebra. Furthermore, $L_{g*} : T_eG \to T_gG$ is a vector space isomorphism and thus not only does $\mathfrak{g}$ always exist (as every group has an identity), but we can identify each tangent space with T_eG. Note that his statement implies that we can identify a Lie algebra with the tangent space since a left-invariant vector field is fully determined by its value at the identity. That is,

Proposition 2.2.9. *The Lie algebra $\mathfrak{g}$ of G can be identified as the tangent space at the identity, T_eG. That is,*

$$\mathfrak{g} \to T_eG$$

such that

$$X \mapsto X_e$$

is a vector isomorphism where X_e denotes the vector field, X at e.

The above proposition implies that we can map curves, $\gamma : I \to G$, in a Lie group G to curves in the corresponding Lie algebra, $\mathfrak{g}$. Explicitly, $X : (0,1) \to \mathfrak{g}$ via $X(t) = dL_{\gamma'(t)}\gamma'(t)$. However, to go in the reverse direction, we require a particular identification of $\mathfrak{g}$. To that end, we define a *one-parameter subgroup* as

Definition 2.2.10. A *one-parameter subgroup* of a Lie group, G is a smooth group homomorphism from $(\mathbb{R}, +)$ to G.

The key point is that the Lie algebra $\mathfrak{g}$ can be identified with the set of all one-parameter subgroups of G as follows.

Proposition 2.2.11. *For every $X \in \mathfrak{g}$, the integral curve of X starting at the identity, $e \in G$, denoted ϕ_X, is a one parameter subgroup of G. Moreover, every one-parameter subgroup of G is of this form.*

With this identification of $\mathfrak{g}$, we can define the *exponential map*, which sends points in $\mathfrak{g}$ to points in G.

Definition 2.2.12. The *exponential map*, denoted exp, is the mapping exp: $\mathfrak{g} \to G$ such that for $X \in \mathfrak{g}$, $\exp(X) = \gamma(1)$ where $\gamma : \mathbb{R} \to G$ is the unique one parameter subgroup of G whose tangent vector at the identity is equal to X.

Note that in the case of matrix Lie groups, the exponential map coincides with the usual matrix exponential. Moreover, notice that in the setting of Definition 2.2.20, the fact that tangent spaces are isomorphic implies that it is sufficient to define a metric on $T_eG \simeq \mathfrak{g}$ to get a left-invariant metric on all of G.

2.2.3 Group actions

Homogeneous spaces involve the actions of Lie groups. As such we begin by recalling the following (standard) definitions from [49].

Definition 2.2.13. Let G be a Lie group acting on M. Then

1. Given a point $p \in M$, we define the *stabiliser* (or *isotropy*) group, $G_p = \{g \in G \mid gp = p\} \subset G$ and the *orbit* $Gp = \{gp \mid g \in G\} \subset M$.

2. The G-action is called *effective* if $N := \cap_{p \in M} G_p = \{e\}$.

3. The G-action is *free* if $G_p = \{e\}$ for all $p \in M$.

4. The G-action is *transitive* if for $p, q \in M$, there exist a $g \in G$ with $gp = q$.

5. The G-action is *proper* if a sequence, $\{p_i\}_{i \in I} \in M$ converges to $x \in M$ and $g_i p_i \to y \in M$, then a subsequence of g_i converges to g with $gx = y$.

Remark 2.2.14. *Note that free actions are necessarily effective but otherwise there are no relations between these concepts in general.*

We can now formally define a homogeneous space. We begin with the following general definition.

Definition 2.2.15. A homogeneous space is a smooth manifold M which is diffeomorphic to a smooth manifold of the form G/H with the quotient topology where G is a Lie group and $H \subset G$ a closed subgroup. Then, G/H has a unique smooth structure such that the quotient map, $\pi : G \to G/H$ is a smooth submersion ([33, Theorem 12]).

Note that G/H in the above definition is a Lie group with its natural group structure. The key theorem for determining whether a space is homogeneous is the following.

Theorem 2.2.16. *Let $L_g : M \to M$ be a transitive left action of a Lie group, G on a smooth manifold, M for some fixed $p \in M$. Let $\pi : G \to G_p$ denote the quotient map and endow G/G_p with the smooth structure of Definition 2.2.15 and define*

$$\varphi : G/H \to M, \quad \phi : \pi(g) \mapsto L_g(p).$$

Then φ is a diffeomorphism and hence M is a homogeneous space.

Theorem (2.2.16) implies that we can equivalently define homogeneous manifolds as smooth manifolds admitting a transitive Lie group action[3]. To put this definition in the language of isometric actions, we recall the following propositions from [49].

Proposition 2.2.17. *If the (left) action of a Lie group, G on M is proper, then all the isotropy groups, G_p are compact and the orbits, Gp are closed in M.*

As a result,

Proposition 2.2.18. *If $G \subset Iso(M)$ is a closed Lie subgroup, then the action of G is proper. In particular, the action of $Iso(M)$ is proper.*

Of course, every action by a compact group is proper and a proper action on a compact manifold, G is compact since $Iso(M)$ is compact. Finally, we note that we can always fix a complete background metric on M which is compatible with the group action if we have compact stabiliser groups.

[3]We note that this is just one of many equivalent ways to define homogeneous manifolds.

Proposition 2.2.19 (Proposition 1.34, [49]). *If G acts properly on M, then there exists a complete Riemannian metric, g on M such that G acts by isometries.*

The proof of this fact uses the slice theorem and tubular neighbourhoods which can be found in [49]. We call a metric that satisfies the hypotheses of (2.2.19) *G-invariant*. In particular, if we suppose that G acts by left multiplication, then we make the following definition.

Definition 2.2.20. Let G be Lie group. Then a Riemannian metric on G is left-invariant if the left multiplication map, $L_g : G \to G$ is an isometry for all $g \in G$.

2.2.4 Representations & invariant metrics

In this section we describe how G-invariant metrics are related to *representations* of groups. Recall that a *representation* of a group, G is a double (V, ρ) where V is a vector space and ρ a group homomorphism, $\rho : G \to GL(V)$. In particular, if V is finite dimensional, we can choose a basis of V to obtain an isomorphism $\mathrm{GL}(V) \simeq \mathrm{GL}_n(V)$. For instance,

Example 2.2.1. Consider $G = \langle g \rangle \simeq \mathbb{Z}_k$. Then, the map $\rho : G \to \mathrm{GL}_1(\mathbb{C}) \simeq \mathbb{C}^\times$ such that $g \to \xi^i$ where ξ is the $n-$th root of unity is a representation of $\mathbb{Z}/n\mathbb{Z}$. Hence, we can identify $\mathbb{Z}_n = \{e^{\frac{2\pi i k}{n}} \mid k = 0, \dots n-1\}$ which defines the action on $\mathbb{C}$. It is immediate that this is indeed a *matrix* Lie group under the identification that matrix Lie groups are subgroups of $GL_n(\mathbb{C})$.

In general, there are many possible representations of a group. For our purposes, a useful representation will be the *adjoint representation*.

Definition 2.2.21. Let G be a Lie group and $\mathfrak{g}$ its associated Lie algebra. let $C_g : G \to G$ be the conjugation map by g. Moreover, let $Ad_g = d(C_g)_e : T_eG \to T_eG$. Then, the Adjoint representation of a group G is the map $Ad : G \to \mathrm{Aut}(\mathfrak{g})$ given by $g \mapsto Ad_g$.

Remark 2.2.22. *Recall that the tangent space is vector space and that $d(C_g)_e \in GL(T_eG)$. Hence, this assignment makes sense.*

One way to pass from a representation of a Lie group, G to a representation of its Lie algebra is by taking the derivative at the identity. Thus, we can also obtain a representation of the corresponding Lie algebra $\mathfrak{g}$ as follows.

Definition 2.2.23. The adjoint representation of $\mathfrak{g}$ is the map $ad : \mathfrak{g} \to \mathrm{End}(\mathfrak{g})$ given by $x \mapsto ad_x = d(Ad_g)_e(X)$. Moreover, $ad_x(y) = [x, y]$.

We can now define the Cartan-Killing form,

Definition 2.2.24. Let $\mathfrak{g}$ be a Lie algebra such that its associated Lie group, G is compact and simple. Then, if $X, Y \in \mathfrak{g}$, Then, the Cartan-Killing form, $B(\cdot, \cdot)$ is given by

$$B(X, Y) = tr(ad_X \circ ad_Y).$$

If we have a basis, $\{u_i\}$ of $\mathfrak{g}$ then the matrix elements of the Killing form are given by

$$B_{ij} = \mathrm{tr}(ad_{u_i} \circ ad_{u_j})$$

The Cartan-Killing form (2.2.24) is connected with invariant metrics as the following proposition shows.

Proposition 2.2.25. *There is a one-to-one correspondence between left-invariant metrics on G and Ad_G-invariant inner products on $\mathfrak{g}$. That is,*

$$\langle Ad_g X, Ad_g Y\rangle_e = \langle X, Y\rangle_e$$

for all $g \in G$.

Clearly, B is Ad_G-invariant and thus defines a left-invariant metric on G. Furthermore, if G is simple and compact, then the left-invariant metrics on G are equivalent to the Cartan-Killing form (2.2.24) up to a rescaling. To make this more concrete, we give an example,

Example 2.2.2. Consider the basis of $\mathfrak{su}(2)$ as a matrix Lie algebra given by the matrices,

$$u_1 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \quad u_2 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \quad u_3 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{2.5}$$

Then the Lie bracket of two elements $u, v \in \mathfrak{g}$ is given by $[u, v] = uv - vu$ [33]. Thus, the basis (2.5) satisfies the following lie bracket relations,

$$[u_i, u_j] = 2u_k$$

for all distinct $i, j, k \in \{1, 2, 3\}$. This basis is known as a *Milnor basis* as studied by Milnor in [34]. We can use these relations and Definition 2.2.24, to define the matrix form B_{ij}. One finds that B is negative definitive and thus $g(u_i, u_j) := -B_{ij}$ defines a bi-invariant metric on $SU(2)$. Furthermore, up to scaling $g(u_i, u_j) = \delta_{ij}$, i.e. g is diagonal.

As shown in Example 2.2.2, the Cartan-Killing form is diagonal on $SU(2)$. In fact, a more general statement is true due to [16].

Lemma 2.2.26. *Let g be any left-invariant metric on $SU(2)$. Then, there exists a standard Milnor basis, $\{e_1, e_2, e_3\}$ that is orthogonal in the metric g and satisfies $g(e_1, e_1) \leq g(e_2, e_2) \leq g(e_3, e_3)$.*

Hence, $SU(2)$ can be equipped with a left-invariant metric that is diagonal in some Milnor basis. To see how this applies to homogeneous spaces, we introduce the *isotropy representation* ([49] [30]).

Definition 2.2.27. Let (M, g) be a homogeneous Riemann manifold and $\mathrm{Iso}_p(M, g)$ denote the group of isometries that fix $p \in M$. Then, (2.2.16) implies that each $\varphi \in \mathrm{Iso}_p(M, g)$ is such that the linear map $d\varphi_p$ is an automorphism of T_pM. Hence the map

$$I_p : \mathrm{Iso}_p(M, g) \to \mathrm{GL}(T_pM)$$

given by

$$\varphi \mapsto d\varphi_p$$

is a representation of $\mathrm{Iso}_p(M, g)$ called the *isotropy representation*.

To phrase Definition 2.2.27 in terms of group actions, suppose that G is Lie group with a left action L_g such that G_p is the isotropy group at the point, p. Then $G/G_p \simeq M$ via the map L_g which is clearly a diffeomorphism. Thus, we can take the derivative, $d(L_g)_p : T_pM \to T_pM$. By the chain rule, it follows that

$$d(L_{gh})_p = d(L_g \circ L_h) = (dL_g)_p \circ (dL_h)_p$$

and thus there is a representation,

$$\rho : G_p \to GL(T_pM)$$

given by $v \mapsto d(L_g)_p(v)$. The concept of an isotropy representation allows us to understand metrics on homogeneous spaces. To make this precise we introduce the concept of *action fields*.

Definition 2.2.28. If $\mathfrak{g}$ is the Lie algebra of G, and $X \in \mathfrak{g}$, then the vector field, X^*, on M satisfies

$$X^*(p) = \frac{d}{dt}\bigg|_{t=0} \exp(tX)p$$

is called the *action field* associated to X.

The notion of action fields allows the algebraic identification of the isotropy representation. That is, if G_p is the stabiliser of G at p, then (according to [49]),

Proposition 2.2.29. *Let $M = G/G_p = G/H$ and $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$ be the decomposition which $Ad_H(\mathfrak{m}) \subset \mathfrak{m}$. Then the linear map, $\mathfrak{m} \to T_pM$ given by $X \to X^*(p)$ is a Ad_H equivariant isomorphism. The isotropy representation of $G_p = H$ on T_pM is thus identified with the action of Ad_H i.e*

$$d(L_g)(X^*(p)) = (Ad(g)(X))^*(p)$$

for $g \in G_p$.

The isomorphism follows from [49, Proposition 1.5] and the fact that M and $\mathfrak{m}$ have the same dimension while the identification follows from [49, Prop 1.5 (c)]. In summary, an adjoint representation is an isotropy representation associated with the conjugation map of G at the identity element of G. To make this more concrete, we given an example.

Example 2.2.3. Consider the matrix Lie algebra, $\mathfrak{su}(2) = \text{span}\{v_1, v_2, v_3\}$ where

$$u_1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \quad u_2 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \tag{2.6}$$

Then for some element $u_\theta \in U(1)$, we have isotropy representations:

$$u_\theta \mapsto Ad_u(v_1) = d(C_u)_e(v_1) = (C_u \circ \exp(tX)')(0) = uv_1u^{-1} = v_1$$

and

$$u_\theta \mapsto \text{Ad}_u(cos(\phi)v_2 + sin(\phi)v_3) = cos(\phi + 2\theta)u_2 + sin(\phi + 2\theta)v_3$$

by straightforward computation ($\text{Ad}_u(v_i) = uv_iu^{-1}$).

We note that the decomposition in (2.2.29) of $\mathfrak{g}$ does not exist in general. However, it is true when H is compact. For example, we can choose an inner product $\langle , \rangle_{\mathfrak{g}}$ on $\mathfrak{g}$ which is Ad_H invariant and thus the orthogonal complement, $\mathfrak{m} = \mathfrak{h}^c$ is invariant under Ad_H.

Note that the isomorphism in Proposition (2.2.29) implies that an Ad_H invariant inner product $\langle , \rangle_{\mathfrak{g}}$ also defines an inner product $\langle , \rangle_M$ on M by left translations via

$$\langle , \rangle_M(g \cdot p) = L_g^*(\langle , \rangle_{\mathfrak{g}})(p) \tag{2.7}$$

which is *independent* of the choice of g since $\langle , \rangle_{\mathfrak{g}}$ is invariant under Ad_H. Conversely, a G-invariant metric on M is *determined* by an Ad_H invariant inner product of $\mathfrak{m}$. Hence we have a one-to-one correspondence between G-invariant metrics on M and Ad_H invariant inner product of $\mathfrak{m}$. Notice also that $\langle , \rangle_{\mathfrak{g}}$ defines a left-invariant metric on G.

We conclude this section by shedding some light on the Ricci solitons that arise from homogeneous manifolds. The [37, No Breathers Theorem] of Perelman implies that on a compact manifold, all Ricci solitons are of gradient type. As a result, Ricci solitons of constant scalar curvature must be Einstein. Explicitly, if we take the trace of (1.6), we get that

$$R + \Delta u + \frac{n}{2}\lambda = 0$$

where R is the scalar curvature and n the dimension of M. Hence if R is constant, we can integrate over M to deduce that $R + n\lambda = 0$ and so u is a constant function, and thus the soliton is Einstein. In other words, *compact* homogeneous solitons are trivial. In contrast, *non-compact* homogeneous Ricci solitons exist and can be non-trivial ([13], [29]).

Indeed, by the work of ([36],[40], [26], [27]), we have the following classification of non-trivial homogeneous Ricci solitons as either

- expanding, non-compact and non-gradient
- shrinking and finitely covered by a product of an Einstein manifold and a shrinking Gaussian.

2.3 Cohomogeneity one manifolds

The next step up in complexity is to manifolds that are foliated by homogeneous spaces of co-dimension one called *cohomogeneity one manifolds*. In this case, we have an $(n+1)$-dimensional connected manifold, M equipped with an action by a compact, connected Lie group, G, such that the orbit space, $M^* := M/G$, is one dimensional. In this case, there are four possibilities up to diffeomorphism [49]:

1. $M^* = \mathbb{R}$,
2. $M^* = [0, \infty)$,
3. $M/G = S^1$,
4. $M/G = [-1, 1]$,

where $(1), (2)$ are non-compact whilst $(3), (4)$ are compact. Note that ([25], [22], [37]), imply that compact Ricci solitons are Einstein or shrinking and of gradient type. We will therefore be interested in the case $M^* = [0, \infty)$.

Remark 2.3.1. *W. Wylie in [48] showed that any complete shrinking Ricci soliton has finite fundamental group and thus $M^* = S^1$ gives rise to trivial Ricci solitons.*

We follow the approach of [49] to describe the topological structure of cohomogeneity one manifolds more concretely. As such, let G_p be the isotropy group of G at $p \in M$. Then observe that stabiliser groups appear in conjugacy classes since $G_{gp} = gG_pg^{-1}$. Hence, we define two points to have the same *isotropy type* if the corresponding isotropy groups are conjugate.

Next, observe that being conjugate to a subgroup of an isotropy group defines a partial order on the set of isotropy types. Explicitly, if K is a stabiliser group of the action by G, then the conjugacy class (K) is given by $(K) := \cup_{g\in G} gKg^{-1}$. We can then define the partial ordering via,

$$(K) \leq (L) \text{ if } gKg^{-1} \subset L \text{ for some } g \in G.$$

In fact, there exists a unique minimal isotropy type called the *principal isotropy group* and the corresponding orbits, G/G_p are called *principal orbits*. The remaining orbits are then classed as *singular* or *exceptional* if they are of dimension greater or equal to the dimension of the principal orbits respectively. Note that the boundary points in the M^* cases above correspond to non-principal orbits.

In this thesis, we are concerned with the second case where the manifold has one singular orbit at $0 \subset [0, \infty) = M^*$, and is non-compact with group diagram $\mathbb{Z}_n \subseteq U(1) \subseteq SU(2)$ as described by W. Ziller in [49] and studied by [43], [41] and [47]. In order to understand these spaces, we need to understand *homogeneous vector bundles*.

2.3.1 Homogeneous vector bundles

The goal of this section is to understand associated bundles which will then allows us to precisely define homogeneous vector bundles. We will see that these are smooth manifolds of the form $M \times_G F$ as in (1.7). We recall our earlier definition of a fibre bundle as in Definition 2.1.2, with an example.

Example 2.3.1. Consider the fibre bundle $\pi : M \times F \to M$ where π is the projection on the first coordinate and M, F are smooth manifolds. In this case, we can take our neighbourhood, U to be all of M and thus define the bundle chart, $\alpha : M \times F \to F$ as the second projection.

We now recall some elementary (but technical) notions related to fibre bundles from differential geometry. Suppose we are given a fibre bundle E over M with fibre F. Then, a *bundle atlas* is a collection of *bundle charts* whose corresponding sets, $U \subset M$ form an open cover of M. Then each fibre, $E_x := \pi^{-1}(x)$ for $x \in M$ is diffeomorphic to F. Since E, M, F are smooth manifolds, we know that if $\alpha : \pi^{-1}(U) \to F$ and $\beta : \pi^{-1}(V) \to F$ are two bundle charts with $U \cap V \neq \emptyset$, then $\alpha|_{\pi^{-1}(U\cap V)}$ and $\beta|_{\pi^{-1}(U\cap V)}$ are also bundle charts. Hence, for each $x \in U \cap V$, both α and β map the fibre E_x diffeomorphically onto F and thus the map,

$$\rho_{\alpha\beta} : U \cap V \to \text{Diff}(F),\ \rho_{\alpha\beta}(x) := \alpha|_{E_x} \circ \beta^{-1}_{E_x}$$

is well defined where $\mathrm{Diff}(F)$ denotes the set of diffeomorphisms on F. We call such a function a *transition function* from the bundle chart α to the bundle chart β. Then the connection between homogeneous spaces and fibre bundles can be illustrated as follows.

Example 2.3.2. Suppose we have a fibre bundle, $\pi : E \to M$ with fibre F. Suppose also that we have a Lie group, G such that the left action, $L : G \times F \to G$ is an effective. Then we say our charts are $(G, L)-$*compatible* if for any two bundle charts, $\alpha : \pi^{-1}(U) \to F$, $\beta : \pi^{-1}(W) \to F$ in a bundle atlas for E such that if $U \cap W \neq \emptyset$, there exists a smooth map, $\tilde{\rho}_{\alpha\beta} : U \cap V \to G$ such that

$$\rho_{\alpha\beta}(x)(z) = L(\tilde{\rho}_{\alpha\beta}(x), z)$$

for $x \in U \cap W$ and $z \in F$ and we say E is a fibre bundle with *structure group* G.

The key point here is that we can define homogeneous spaces as fibre bundles. For instance,

Example 2.3.3. Let G be a Lie group and $H \subset G$ a closed subgroup. Then,

$$\pi : G \to G/H$$

is a fibre bundle with structure group H where H acts on itself via left translations. Note that G/H is indeed a topological manifold with the quotient topology and such that $\pi : G \to G/H$ is a submersion [33, Theorem 12.4] which we refer to as the *smooth structure*.

We now give some examples of specific types of fibre bundles. The first is a *vector bundles* where the Lie group is simply a matrix group. Indeed,

Definition 2.3.2. A *vector bundle of rank* n is a fibre bundle, $\pi : E \to M$ with fibre $F = \mathbb{R}^n$ and structure group $G = GL_n(k)$ or some matrix Lie subgroup thereof where k is a field.

We now consider fibre bundles where the fibre, F is a Lie group that acts freely on the total space. We call these objects *principal G-bundles*.

Definition 2.3.3. Let $\pi : P \to M$ be a fibre bundle with fibre a Lie group G. Assume that there exists a free, fibre-preserving right action of G on P and a bundle atlas for P with the property that each bundle chart $\alpha : \pi^{-1}(U) \to G$ is *G-equivariant* in the sense that

$$\alpha(p \cdot g) = \alpha(p)g$$

for all $p \in \pi^{-1}(U)$, $g \in G$. Then P is a *principal G-bundle* over M with group G.

The simplest example is the bundle $\mathrm{Pr}_1 : M \times G \to M$ where the action is given by $(x, g) \cdot h := (x, gh)$ (see [33] for an explicit computation). We now state the following theorem that allows us to construct more examples of principal bundles [33, Section 12].

Theorem 2.3.4. *Let P be a manifold and G a Lie group acting freely and properly. Then P/G admits a unique smooth structure such that the projection is a surjective submersion and P/G has the quotient topology.*

Corollary 2.3.4.1. *Let P be a manifold and G a lie group acting freely and properly, then $\pi : P \to P/G$ is a principal $G-$bundle.*

We can now build *associated bundles* which are our fundamental object of interest in the construction of cohomogeneity one manifolds.

Definition 2.3.5. Let $\pi : P \to M$ be a a principal G-bundle and assume that G acts freely on another manifold, F via the left action, $L_g : G \times F \to F$. Define an equivalence relation $\sim$ on $P \times F$ via

$$(p \cdot g, v) \sim (p, L(g, v))$$

for all $p \in P$, $g \in G$, $v \in F$. Then we denote the quotient space $(P \times F)/\sim$ as $P \times_G F$. Note that by writing $[p, x]$ for the equivalence classes, we can define a map, $\Phi : P \times_G F \to M$ by $\Phi : [p, x] \mapsto \pi(p)$.

Note that the notation $P \times_G F$ is ambiguous as the action of G is not defined. Thus unless stated otherwise, assume that the action is always on the left. Then the following theorem gives us all the information we need about associated bundles.

Theorem 2.3.6 (Theorem 25.3, [33])**.** *Let $\pi : P \to M$ be a principal G-bundle and assume G acts freely on a manifold, F, as a left action, $L_g : G \times F \to F$. Then*

1. *$P \times_G F$ is a smooth manifold.*
2. *$\Phi : P \times_G F \to M$ is a fibre bundle with fibre F and structure group G.*
3. *P is the principal bundle associated to $P \times_G F$.*
4. *The quotient map $q : P \times F \to P \to P \times_G F$, given by $q(p, v) := [p, v]$ is a principal G-bundle.*
5. *For each $p \in P$, the map $R_p : F \to P \times_G F$ given by $v \mapsto [p, v]$ is a diffeomorphism over the fibre $\pi(p)$.*
6. *If F is a vector space and G acts linearly on F. Then L_g is a representation $G \to GL(F)$ and $P \times_G F$ is a vector bundle over M and R_p from the previous part is a linear isomorphism.*

Then our manifold of interest, $SU(2) \times_{U(1)} \mathbb{R}^2$ is the associated vector bundle described in the final point of Theorem 2.3.6 which we call a *homogeneous vector bundle*.

2.3.2 Topological structure

By the previous discussion, our manifold of interest,

$$M = SU(2) \times_{U(1)} \mathbb{R}^2$$

given by the group diagram $\mathbb{Z}_n \subseteq U(1) \subseteq SU(2)$ is the homogeneous vector bundle as in the final point of Theorem 2.3.6. In this section, we carefully describe the topological structure of this manifold.

As M satisfies the hypotheses of (2.3.6), we can consider $\mathbb{Z}_k \subset U(1) \subset SU(2)$ to be *matrix* Lie groups given by some representations. To this end, recall that $SU(2) \in \mathrm{GL}_2(\mathbb{C})$ is given by,

$$SU(2) = \left\{ \begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix} \;\middle|\; \alpha, \beta \in \mathbb{C},\ |\alpha|^2 + |\beta|^2 = 1 \right\}$$

which has closed subgroups,

$$\mathbb{Z}_k = \left\{ \begin{pmatrix} e^{2\pi i \frac{k}{n}} & 0 \\ 0 & e^{-2\pi i \frac{k}{n}} \end{pmatrix} \;\middle|\; k \in \{0, 1, \ldots, n-1\} \right\} \subset U(1) = \left\{ \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix} \;\middle|\; \theta \in [0, 2\pi) \right\} \subset SU(2).$$

Clearly, one can define a *surjective* map $\varphi : U(1) \to \mathbb{S}^1$ such that $\ker(\varphi) = \mathbb{Z}_k$. Hence, $U(1)/\mathbb{Z}_k \simeq \mathbb{S}^1$. We note that this action extends to a linear action on $\mathbb{R}^2$ that preserves and acts transitively on circles centered at the origin. Hence, by (2.3.6), we have an equivalence relation $[g, x] \sim [gk^{-1}, kx]$ for $g \in SU(2)$, $k \in U(1)$, and $x \in \mathbb{R}^2$.

We now define an action of $SU(2)$ on M via

$$(h, [g, x]) \mapsto [gh, x]$$

where $h \in SU(2)$. Note that this action is well defined as $SU(2)$ acts transitively on itself. Note that this construction implies that the action of $\mathbb{Z}_k$ satisfies

$$\begin{aligned} z[g, x] &= [gz, x] \\ &= [g, xz] = [g, x]. \end{aligned}$$

That is, $\mathbb{Z}_k$ is the isotropy group for every point in M for $x \neq 0$. Clearly, if $x = 0$ then the isotropy group is $U(1)$. Since $\mathbb{Z}_k \in Z(SU(2))$, it follows that

$$g\mathbb{Z}_n g^{-1} = \mathbb{Z}_n \leqslant U(1)$$

and thus $(\mathbb{Z}_n) \subseteq (U(1))$ according to the partial ordering defined in [49]. Hence $\mathbb{Z}_k$ is the *principal isotropy* group and thus the principal orbits are given by $SU(2)/\mathbb{Z}_k$ on $(0, \infty) \subset M^*$ where $M^* = M/G = [0, \infty)$ whilst $SU(2)/U(1)$ is the (unique) singular orbit at $x = 0$ by definition. Finally, notice that $\mathbb{R}^2/U(1) \simeq [0, \infty)$. Hence, we can identify the *principal part*, $M_0 \subset M$ topologically as $M_0 := I \times SU(2)/\mathbb{Z}_k$ where $I = (0, 1)$ and a *singular part*, $\{0\} \times SU(2)/U(1) \subset M$.

Remark 2.3.7. *For ease of notation, we will sometimes write* (H, K, G) *instead of* $(\mathbb{Z}_k, U(1), SU(2))$ *respectively.*

2.3.3 Geometric structure

We now describe the geometric structure of M in detail. By Proposition 2.2.19, we can assume that there exists a complete, G-invariant metric, g on M.

Following the approach of [43], [41], we choose a normal geodesic $\gamma : (0, 1) \to M$ perpendicular to all principal orbits. Recall that $\mathbb{Z}_k$ is the principal isotropy group and is thus constant along

γ [49, Lemma 1.38] (also known as Kleiner's Lemma). As a result, there is a G-equivariant diffeomorphism,

$$\Phi : I \times SU(2)/\mathbb{Z}_k \to M_0$$

given by

$$(r, gH) \mapsto g \cdot \gamma(r)$$

where $M_0 \subset M$ is the open, dense subset consisting of points lying on principal orbits and $I = (0, 1)$. Hence, the metric $\Phi^* g$ on $I \times SU(2)/\mathbb{Z}_k$ is of the form

$$\Phi^* g = dr^2 + g_r \tag{2.8}$$

where $\{g_r\}_{r \in I}$ is a one-parameter family of $SU(2)$-invariant Riemannian metrics on $SU(2)/\mathbb{Z}_k$.

To understand the metrics on the homogeneous space, $SU(2)/\mathbb{Z}_n$, recall that if $\tilde{g}$ is our Ad_G-invariant inner product on $\mathfrak{g}$ (as in Example 2.2.2), then we can define a $\tilde{g}$-symmetric endomorphism, P_r, on $\mathfrak{m}$ (as in Proposition 2.2.29) such that $\tilde{g}(P_r \cdot, \cdot)$ defines an Ad_H-invariant inner product on $\mathfrak{m}$ and thus induces a G-invariant inner product on G/H. Indeed, T. Buttsworth in [43, Proposition 1] used a diagonalising Milnor frame to derive a $SU(2)$ invariant metric, g_{r_0} on $SU(2)/\mathbb{Z}_k$ as follows.

Proposition 2.3.8. *Let g be a left-invariant Riemannian metric on $SU(2)$. Then there are left invariant vector fields, $\{X_i\}_{i=1}^3$ satisfying*

$$[X_i, X_j] = X_k$$

for all $i, j, k \in \{1, 2, 3\}$ distinct, such that

$$g = f_1^2 \omega_1^2 + f_2^2 \omega_2^2 + f_3^2 \omega_3^2 \tag{2.9}$$

for positive numbers f_i and ω_i the one-forms corresponding to the left-invariant vector fields X_i. This metric passes to an $SU(2)$ invariant Riemannian metric on $SU(2)/\mathbb{Z}_k$ if and only if one of the following holds:

- *$k = 1, 2$;*
- *$k = 4$ and the frame $\{X_i\}_{i=1}^3$ is given by $\{\frac{u_1}{2}, A\frac{u_2}{2}, A\frac{u_3}{2}\}$ for $A \in SO(2)$ respectively,*
- *$k \notin \{1, 2, 4\}$ and our diagonalising frame has $X_i = \frac{u_i}{2}$ for u_i defined in (2.6) and $f_2 = f_3$.*

Proof. Recall that g as in (2.2.2) is a left-invariant Riemann metric on $SU(2)$. Then we have left invariant vector fields X_i given by $\{\frac{u_1}{2}, \frac{u_2}{2}, \frac{u_3}{2}\}$ as in (2.6), which satisfy

$$[X_1, X_2] = X_3, \quad [X_2, X_3] = X_1, \quad [X_3, X_1] = X_2$$

called a *Milnor Frame* [34]. This frame gives rise to a diagonal metric (cf. [16],[34])

$$g = f_1^2 \omega_1^2 + f_2^2 \omega_2^2 + f_3^2 \omega_3^2$$

on $SU(2)$. To pass to a metric on $SU(2)/\mathbb{Z}_n$ we use Proposition 2.2.29. That is, the metric (2.9) passes to an $SU(2)$-invariant metric only if it is $Ad_{\mathbb{Z}_k}$ invariant. Precisely, for $u_\theta \in \mathbb{Z}_k$, the relation

$$\langle Ad_{u_\theta} u_i, Ad_{u\theta} u_j \rangle = \langle u_i, u_j \rangle$$

must hold. For $k = 1, 2$, Ad_{u_θ} acts as the identity for each $u_\theta \in \mathbb{Z}_k$ and thus all metrics of the form (2.9) are invariant in the quotient. However, if $k = 4$, then we can set $\phi = 0$ with respect to the relations in (2.2.3) such that,

$$Ad_{u_\theta}(u_2) = \cos(2\theta)u_2 + \sin(2\theta)u_3.$$

Thus if $k = 1$, then $Ad_{u_\theta}(u_2) = -v_2$ and similarly, if $k = 2$, then $Ad_{u_\theta}(u_2) = u_2$. Thus two elements, $u_\theta \in \mathbb{Z}_k$ act as the identity and two as

$$Ad_{u_\theta}(u_1) = u_1, \quad Ad_{u_\theta}(u_2) = -u_2, \quad Ad_{u_\theta}(u_3) = -u_3,$$

As a result, there is a $u_\theta \in \mathbb{Z}_k$ such that

$$\langle Ad_{u_\theta} u_1, Ad_{u_\theta} u_2 \rangle = \langle u_1, -u_2 \rangle$$

which implies that $\langle u_1, u_2 \rangle = 0$ and similarly for $\langle u_1, u_3 \rangle$. Hence, the metric is invariant if and only if u_1 is orthogonal to u_2 and u_3. By applying a rotation to the $\{u_2, u_3\}$ basis, we obtain a Milnor frame and the assertion follows.

For $k \notin \{1, 2, 4\}$ recall that $Ad_{u_\theta}(u_1) = u_1$ for every $u_\theta \in \mathbb{Z}_k$. Moreover, by setting $\phi = 0$ and $\phi = \frac{\pi}{2}$ alternately, we can find $u_\theta \in \mathbb{Z}_k$ such that $Ad_{u_\theta}(u_i) \neq u_i$ for both $i \in \{2, 3\}$. Hence, Ad_{u_θ} invariance holds if and only if u_1 is orthogonal to u_2 and u_3 and that the metric restricted to the $\{u_2, u_3\}$ basis is simply $g = f_{ij}\delta_{ij}$ for $f_{ij} \in \mathbb{R}_{>0}$. Then, we get a Milnor frame diagonalising g such that $f_2 = f_3$ as required. □

We claim that $f_2 = f_3$ implies that the metric g_{r_0} in (2.9) is $U(2)$-invariant under a suitable action.

Lemma 2.3.9. *Suppose that g is a Riemannian metric on $SU(2)$ of the form (2.9). Then g is $U(2)$ invariant if and only if $f_2 = f_3$ under the action of $U(2)$ defined by*

$$\phi : U(2) \times SU(2) \to SU(2)$$

such that

$$\phi : g \mapsto h u_\theta g u_\theta^{-1}$$

for $g, h \in SU(2)$ and $u_\theta \in U(1)$ as matrix multiplication.

Proof. Recall that $U(2) \simeq SU(2) \rtimes U(1)$. As a result, the vector fields $\{X_i\}_{i=1}$ on $SU(2)$ as in Proposition 2.3.8, satisfy

$$\phi_* X_1 = X_1, \quad \phi_* X_2 = \cos(\theta)X_2 - \sin(\theta)X_3, \quad \phi_* X_2 = \cos(\theta)X_2 + \sin(\theta)X_3$$

then, $U(2)$-invariance requires $(\phi_{h,u_\theta})^*(X_i, X_j) = (X_i, X_j)$ which implies the assertion by simple computation. □

It follows from Proposition 2.3.8 and Lemma 2.3.9 that,

Corollary 2.3.9.1. *The metric g of the form (2.9) defined on the homogeneous space, $SU(2)/\mathbb{Z}_k$ is $U(2)$-invariant if and only if $f_2 = f_3$.*

Remark 2.3.10. *This same argument holds on $I \times SU(2)$ and therefore on $I \times SU(2)/\mathbb{Z}_k$ by defining the action $(h,k) \in SU(2) \times U(1)$ corresponding to a element of $U(2)$ as matrix multiplication via $(h,k)(r,g) \mapsto (r, hkgh^{-1})$.*

We now want to extend this frame over $I \times SU(2)/\mathbb{Z}_n$. Observe that the Ad_H-invariant inner product restricted to an Ad_H-irreducible subspace is uniquely determined up to scalings[4]. Thus we wish to decompose $\mathfrak{m} = \mathfrak{h}^\perp$ into mutually orthogonal Ad_H-irreducible subspaces $\mathfrak{m} = \mathfrak{m}_1 + \ldots + \mathfrak{m}_n$ such that there exist smooth, scalar functions, $f_i : I \to \mathbb{R}_{>0}$ satisfying

$$g_r|_{\mathfrak{m}_i} = f_i(r)^2 Q|_{\mathfrak{m}_i}.$$

We now prove a proposition given by T. Buttsworth in [43], that shows that such functions exist and define a metric over $I \times SU(2)/\mathbb{Z}_n$.

Proposition 2.3.11. *For $k \geq 3$, there is a $U(2)$-invariant Riemannian metric of the form (2.8), on $I \times SU(2)/\mathbb{Z}_k$ which is a gradient Ricci soliton with an $SU(2)$-invariant potential function u, there is basis of vector fields $\{X_1, X_2, X_3\}$ on $SU(2)$ such that $[X_i, X_j] = X_k$ for all $i,j,k \in \{1,2,3\}$ distinct such that the metric is of the form*

$$g = dr^2 + f_1(r)^2\omega_1^2 + f_2(r)^2\omega_2^2 + f_3(r)^2\omega_3^2$$

where $f_i : I \to \mathbb{R}_{>0}$ are smooth functions such that $f_2 = f_3$ uniformly.

Proof. Recall that the $SU(2)$ action has the principal isotropy group, $H = \mathbb{Z}_k$. Thus $\mathfrak{m} = \mathfrak{h}^\perp \subseteq \mathfrak{g}$ is the Ad_H-invariant complement of $\mathfrak{m}$ (Since H is compact). We observe that $\mathfrak{m} = \mathfrak{su}(2)$. Now for fixed $r \in (0,1)$, we know from Proposition 2.3.8 that there exists a basis of vector fields $\{X_i\}$ that define an $SU(2)$-invariant metric on $SU(2)/\mathbb{Z}_k$. Hence we can identify $Q(P_r\cdot,\cdot)$ as the matrix $(Q(P_rX_i, X_j))_{i,j}$. Since $Q(P_r\cdot,\cdot)$ is symmetric, it can be diagonalised by applying an $SO(2)$ action to the basis elements such that the conditions continue to hold. Thus, we can suppose we have a diagonal basis $\{X_i\}_{i=1}^3$ of $Q(P_r\cdot,\cdot)$ that satisfies the necessary conditions to get a frame of action fields, $\{X_i^*\}_{i=1}^3$ for g_r. Suppose $X \in \text{span}\{X_i^*\}$. Then, by [3],

$$\text{Ric}(X^*, \partial_r) = \frac{1}{2} g_r^{ij} \partial_r (g_r(X_i^*, [X_j^*, X^*])). \tag{2.10}$$

To simplify this expression, recall that the potential function $u : I \to \mathbb{R}$ is $SU(2)$-invariant. Hence, $\text{Hess}(u)(\partial_r, X^*) = 0$. Thus, (1.6) implies

$$\text{Ric}(X^*, \partial_r) = -\text{Hess}(u)(\partial_r, X^*) = 0$$

Hence, we can simplify Equation (2.10) to

$$(g_r^{ii} - g_r^{jj}) g'_{r_{ij}} = 0$$

[4]This follows from the discussion in Example 2.2.2.

which shows that g_r is diagonal (by applying an $SO(2)$ action if necessary). Hence, for fixed $r \in I$, $\{\partial_r^*, X_i^*\}$ is a diagonalising frame on a tubular neighbourhood of $G \cdot \gamma(r)$ such that both $g_{\gamma(r)}$ and $g'_{\gamma(r)}$ are diagonal. According to [41] (for instance), the other gradient Ricci soliton equations become,

$$-tr(L') - tr(L^2) + u'' - \lambda = 0 \tag{2.11}$$
$$R(g) - tr(L)L - L' + u'L - \lambda I = 0 \tag{2.12}$$
$$g' - 2gL = 0 \tag{2.13}$$

where $R(g) : \mathfrak{su}(2) \to \mathfrak{su}(2)$ is the Ricci curvature operator of $g(r)$ on $SU(2)$ and $L(r)$ the shape operator of the Riemann submanifold $(SU(2), g_r)$ (defined implicitly from the third equation of (2.11)). Since g_r, g'_r are diagonal at r, the equations (2.12, 2.13) imply that L,L' are also diagonal. Hence, for any fixed $r \in (0, R)$, the existence and uniqueness of solutions to systems of ODEs imply that the metric is diagonal for all $r \in (0, R)$.

□

As we are concerned only with $U(2)$-invariance, we assume from now on that $f_2 = f_3$ uniformly for all $k \in \mathbb{N}$.

Remark 2.3.12. *We refer the interested reader to a paper by A. Dancer in [41] which introduced a formalisation for this construction.*

Remark 2.3.13. *Note that by choosing our basis in this way, we will obtain a slightly different expression for the Ricci curvature relative to Appleton in [1].*

Chapter 3

$U(2)$-invariant steady Ricci solitons

In this section, we will first note some consequences of the equations (2.11)-(2.13) before turning our attention to solving (2.11) in terms of the functions $f_i : (0, \infty) \to \mathbb{R}_{>0}$. First we give a brief overview of the *essential* quantities that authors studying non-compact, Ricci solitons exploit. However, we specialise this to the steady case. In [11], the authors found the following equality

$$u'' + (-u' + tr(L))u' = C$$

where C is some constant which refer to as the *conversation law*. Furthermore, if we subtract the trace of (2.13) from (2.11), we obtain

$$u'' - u'tr(L) + tr(L)^2 - tr(L^2) - tr(R(g)) = 0$$

which implies that the soliton potential function, u is indeed determined by the metric up to constants since the equation is linear in u'. The quantity, $-u + tr(L)$ is of fundamental importance in the analysis of solitons of this type. As such we define

$$\xi := -u' + tr(L)$$

Remark 3.0.1. *ξ can be intuitively understood as a generalisation of the hypersurface mean curvature [41].*

Remark 3.0.2. *In our case, $tr(L) = \frac{f_1'}{f_1} + 2\frac{f_2'}{f_2}$.*

Moreover, the authors of [15] proved that the quantity $(-\xi)^{-2}$ is a Lyapunov function. This connection in tandem with the conservation law was used in [43] to analyse trajectories of the (2.11) system, by reformulating the problem in terms of center manifolds. Finally, this conservation law allows the derivation of an equality for the scalar curvature, R,

$$R + (u')^2 = -C. \tag{3.1}$$

This equality was used by A. Appleton in [1] to conclude that $\Delta u - (u')^2 = R(0) = -2u''(0)$. In Section 3.3 we will see that the soliton equations depend on $-u''(0)$ and thus (3.1) was a crucial in establishing monotonic properties of u in [1]. Furthermore, Chen in [14] proved that $R \geq 0$ for complete, steady solitons, with equality if and only if g is Ricci flat. Hence, if a solution is non-trivial, then $C < 0$.

3.1 Ricci soliton equation

We now derive the Ricci soliton equations in terms of the smooth functions, $f_i : (0,\infty) \to (0,\infty)$ described in Section 2.3.3. Recall that steady, gradient Ricci solitons satisfy the equation

$$\mathrm{Ric}(g) + \mathrm{Hess}(u) = 0$$

where $u : I \times SU(2)/U(1) \to \mathbb{R}$ is a $SU(2)$-invariant potential function [41, Section 2].

Proposition 3.1.1. *The only non-zero terms of the Hessian matrix of u are*

$$Hess(\partial/\partial_r, \partial/\partial_r) = u''$$
$$Hess(X_i, X_i) = \frac{f_i'}{f_i}u'$$

for $i \in \{1,2,3\}$.

Proof. The scalar potential function, $u : I \times SU(2)/U(1) \to \mathbb{R}$ depends only on $r \in I$. That is, $u(r) = u(r, U(1))$ as u is $SU(2)$-invariant. Furthermore, the vector field, X in the definition of the Ricci soliton equation satisfies $X = \mathrm{grad}\ u$. Hence,

$$\mathrm{Hess}(X_i, X_j)(u) = \langle \nabla_{X_i} \mathrm{grad}\ u, X_j \rangle$$

and we can use Koszul's formula (2.3) to get that the only non-zero terms are

$$\mathrm{Hess}(\partial_r, \partial_r) = u''$$
$$\mathrm{Hess}(X_i, X_i) = \frac{f_i'}{f_i}u'$$

for $i \in \{1,2,3\}$.

□

To derive the Ricci curvature tensor, we simply apply [20, Proposition 1.14] as follows.

Lemma 3.1.2. *The non-zero terms of the Ricci curvature tensor for $k \notin \{1,2,4\}$ satisfy:*

$$Ric(\partial_s, \partial_s) = -\frac{f_1''}{f_1} - 2\frac{f_2''}{f_2}$$
$$Ric(u_1, u_1) = \frac{f_1''}{f_1} + 2\left(\frac{f_1^3}{4f_2^4} - \frac{f_1'f_2'}{f_2}\right)$$
$$Ric(u_i, u_i) = \frac{f_2''}{f_2} + \frac{1}{f_2} - \frac{f_1^2}{2f_2^3} - \frac{f_1'f_2'}{f_1} - \frac{(f_2')^2}{f_2}$$

for $i \in \{2,3\}$ where $Ric(u_2, u_2) = Ric(u_3, u_3)$.

Proof. We can uniquely decompose $\mathfrak{n} = \mathfrak{h}^\perp$ into Ad_H irreducible subspaces $\mathfrak{n} = \mathfrak{n}_1 + \mathfrak{n}_2$ given by $\mathfrak{n}_1 = \mathrm{span}\{\frac{u_1}{2}\}$ and $\mathfrak{n}_2 = \mathrm{span}\{\frac{u_2}{2}, \frac{u_3}{2}\}$. Thus, Ric is diagonal on $\{\partial_r, \frac{u_1}{2}, \frac{u_2}{2}, \frac{u_3}{2}\}$ and the statement follows by direct application of [20, Proposition 1.14]. □

3.2 Smoothness at the singular orbit

To smoothly extend the metric defined on $I \times SU(2)/\mathbb{Z}_k$ (as in 2.3.11) over the singular orbit at $r = 0$, we use the conditions given in [45] for all $k \in \mathbb{N}$.

Theorem 3.2.1. *The metric in Proposition 2.3.11 defined on $(0, \infty) \times SU(2)/\mathbb{Z}_k$ is smoothly extendable to the singular orbit at $r = 0$ if and only if there are smooth functions, $\phi_1, \phi_2, \phi_3 : \mathbb{R} \to \mathbb{R}$ with $\phi_2(0) > 0$ such that*

$$f_1(r)^2 = \frac{k^2 r^2}{4} + r^4 \phi_1(r^2), \quad f_2(r)^2 = \phi_2(r^2) + r^{\frac{4}{k}} \phi_3(r^2), \quad f_3(r)^2 = \phi_2(r^2) - r^{\frac{4}{k}} \phi_3(r^2) \quad (3.2)$$

for $k \in \mathbb{N}$.

Remark 3.2.2. *Clearly, if $k \in \mathbb{N} \backslash \{1, 2, 4\}$, then ϕ_3 vanishes as $f_2 = f_3$ uniformly (2.3.11) and the expressions (3.2.1).*

Using (3.2.1), we can now derive the boundary conditions at $r = 0$.

Proposition 3.2.3. *The boundary conditions for f_i at $r = 0$ satisfy:*

$$f_1(0) = 0, \quad f_1'(0) = \frac{k}{2}, \quad f_2(0) = \overline{f_2} > 0, \quad f_2'(0) = 0.$$

Proof. Clearly $f_1(0) = 0$ and $f_2(0) > 0$. Then $f_2(0)' = 0$ by the chain rule. Let $q(r) := f_1(r)^2$. Then, $q' = 2 f_1(r) f_1(r)'$ and $q'(0) = 0$, $q''(0) = \frac{k^2}{2}$. Hence, by L'Hôpital's rule,

$$f_1'(0) = \lim_{r \to 0} f_1'(r) = \lim_{r \to 0} \frac{q'(r)}{2 f_1(r)} = \lim_{r \to 0} \frac{q''(r)}{2 f_1'(r)}$$

and thus $f_1'(0) = \frac{k}{2}$. □

Finally, the existence of the singular orbit at $r = 0$ imposes the smoothness condition, $u'(0) = 0$ on the soliton potential [47]. We note that the Ricci soliton equation is invariant under changing the potential by an additive constant [44]. Thus, we can also assume that $u(0) = 0$. In summary, the boundary conditions at $r = 0$ read:

$$\begin{aligned} u(0) = 0, \quad & u'(0) = 0, \\ f_1(0) = 0, \quad & f_1'(0) = \frac{k}{2}, \\ f_2(0) > 0, \quad & f_2'(0) = 0. \end{aligned}$$

3.3 The initial value problem

From the work of the previous section, we see that the metric satisfies the steady gradient Ricci solitons equations for $k \notin \{1, 2, 4\}$ if and only if the following ordinary differential equations hold:

$$u'' = \frac{f_1''}{f_1} + 2\frac{f_2''}{f_2} \tag{3.3}$$

$$f_1'' = 2\left(\frac{f_1^3}{4f_2^4} - \frac{f_1'f_2'}{f_2}\right) + f_1'u' \tag{3.4}$$

$$f_2'' = \frac{1}{f_2} - \frac{f_1^2}{2f_2^3} - \frac{f_1'f_2'}{f_1} - \frac{(f_2')^2}{f_2} + f_2'u'. \tag{3.5}$$

Notice that the soliton equations (3.3)-(3.5) with the above boundary conditions are degenerate at $r = 0$. For instance, $\frac{f_1'}{f_1}$ becomes unbounded. It turns out that we must specify $u''(0)$ to obtain a unique solution. We claim that the solutions depend continuously on $\alpha := -u''(0)$ and $f_2(0)$. To this end, consider the change of variables,

$$\xi = \frac{f_1'}{f_1} + 2\frac{f_2'}{f_2} - u',\ L_1 = \frac{f_1'}{f_1}$$
$$R_1 = \frac{f_1}{f_2^2},\ R_2 = \frac{1}{f_1},\ L_2 = \frac{f_2'}{f_2}$$

Then, the soliton equations (3.3)-(3.5) become,

$$\xi' = -L_1^2 - 2L_2^2 \tag{3.6}$$

$$L_1' = -\xi L_1 + \frac{R_1^2}{2} \tag{3.7}$$

$$L_2' = -\xi L_2 + \frac{R_2^2}{2} - \frac{(R_1 - R_2)^2}{2} \tag{3.8}$$

$$R_1' = R_1(L_1 - 2L_2) \tag{3.9}$$

$$R_2' = -R_2L_1. \tag{3.10}$$

This change of variables will also be useful in establishing the short term existence of solutions to (3.3)-(3.5). Indeed, T. Buttsworth in [43] uses this change of variables to establish the following theorem concerning $U(2)$-invariant solutions.

Theorem 3.3.1 (Theorem 5, [43])**.** *For each $k \in \mathbb{N}\backslash\{1, 2, 4\}$ there exists a continuous function $\phi : \mathbb{S}^1 \to (1, \infty) \times \mathbb{R}^5$ such that:*

- *there is a $T > 0$ such that for each $(\alpha, \beta) \in \mathbb{S}^1$, there is solution (ξ, L_i, R_i) of (3.6)-(3.10) on $(0, T]$ such that $(\xi, L_i, R_i)(T) = \phi(\alpha, \beta)$ and $\xi(T) > 1$ for $i \in \{1, 2\}$.*
- *the solution satisfies*

$$\alpha = -u''(0) = \lim_{s\to 0} \frac{\xi(r) - L_1(r) - 2L_2(r)}{r}, \quad \beta = \lim_{r\to 0} \frac{R_1(r)}{r}.$$

- *if $\beta > 0$, then $R_1 \equiv 0$ and if $\beta > 0$, then the corresponding solution of (3.3) satisfies the smoothness conditions [45], at $r = 0$.*

Before we come to the proof, we note that,

Lemma 3.3.2. *We can assume that the solutions are characterised by one parameter, $\gamma \in \mathbb{R}$ where $\gamma := -u''(0)$. Explicitly, for $(\alpha, \beta) \in \mathbb{S}^1$,*

$$\gamma := \frac{k\alpha}{2\beta}. \tag{3.11}$$

Proof. Notice that $(\alpha, \beta) \in S^1$. Hence, we can project (α, β) onto the line $\beta = \frac{k}{2}$ and apply the scaling symmetry $g \mapsto \delta g$ for $\delta \in \mathbb{R}$. □

We would like to give some more detail regarding the existence and uniqueness of solutions to (3.6)-(3.6). In particular, we illustrate how one constructs a formal power series solution that satisfies the first and second points of Theorem 3.3.1.

Following the approach of [43], we look for solutions of the form

$$\xi = \frac{1}{r} + \eta_0 \quad L_1 = \frac{1}{r} + \eta_1 \quad L_2 = \eta_2 \quad R_1 = \eta_3 \quad R_2 = \frac{1}{2r} + \eta_4$$

where η_i are smooth functions satisfying $\eta_i(0) = 0$. Thus, the equation for $\eta = (\eta_0, \dots, \eta_4)$ becomes

$$\eta'(r) = \frac{A\eta(r)}{r} + B(\eta(r))$$

where

$$A = \begin{pmatrix} 0 & -2 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & -1/2 & 0 & 0 & -1 \end{pmatrix}.$$

and B is smooth with $B(0) = 0$. Note that A is diagonalisable as $A = PDP^{-1}$ where

$$D = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -2 \end{pmatrix}.$$

Then with $\tilde{\eta} = P^{-1}\eta$ we get

$$\tilde{\eta}' = \frac{D\tilde{\eta}}{r} + P^{-1}B(P\tilde{\eta}) \tag{3.12}$$

Suppose $\tilde{\eta} = \sum_{n\geq 0} \frac{a_n r^n}{n!}$, is a power series solution for $a_n \in \mathbb{R}^5$. Then $\eta' = \sum_{n\geq 1} \frac{a_n r^{n-1}}{(n-1)!}$, and $P^{-1}B(P\eta) = \sum_{n=0} \frac{B_n r^n}{n!}$ since B is analytic in some neighbourhood about $r = 0$.
Thus, (3.12) becomes

$$\sum_{n\geq 1} \frac{a_n r^{n-1}}{(n-1)!} - \frac{D}{r} \sum_{n\geq 0} \frac{a_n r^n}{n!} - \sum_{n\geq 0} \frac{B_n r^n}{n!} = 0.$$

Then multiplication by r implies,

$$\sum_{n\geq 1}\frac{a_n r^n}{(n-1)!} - D\sum_{n\geq 0} a_n\frac{r^n}{n!} - \sum_{n\geq 0}\frac{B_n r^{n+1}}{n!} = 0.$$

By adjusting summation indices, we find

$$\sum_{k\geq 1}\frac{a_k r^k}{(k-1)!} - D\sum_{k\geq 0}\frac{a_k r^k}{k!} - \sum_{k\geq 1}\frac{B_{k-1} r^k}{(k-1)!} = 0$$

and thus we conclude,

$$-Da_0 + a_1 r - Da_1 r - B_0 r + \sum_{k\geq 2}\left[\frac{a_k}{(k-1)!} - D\frac{a_k}{k!} - \frac{B_{k-1}}{(k-1)!}\right] r^k = 0.$$

Recall $B(0) = 0$ and thus $B_0 = 0$. Hence, we have the conditions

$$\begin{aligned} a_0 &= 0 \\ a_1 - Da_1 &= 0 \\ ka_k - Da_k &= kB_{k-1} \end{aligned}$$

We see that a solution will have $a_0 = 0$ and $a_1 = 0$ except for the third and fourth coordinates which are arbitrary. Note that we can interpret our power series solution as the Taylor series solution about $r = 0$, hence $a_0 = \tilde{\eta}(0) = 0, a_1 = \tilde{\eta}'(0)$. For our purposes, we set $\tilde{\eta}_3'(0) = \beta$ and vary $\tilde{\eta}_4'(0)$ uniquely to satisfy the relation on α. With respect to [17], the truncation to the 2nd order formal power series solution then gives rise to a real $U(2)$-invariant metric on the interval $(0, T]$ by [32, Theorem 7.1]. The uniqueness of g then follows from a Picard Iteration argument given in [17].

With the short term existence of this metric established, we take the change of variables,

$$X_i = \frac{1}{-u' + tr(L)}\frac{f_i'}{f_i}, \quad Y_1 = \frac{1}{-u' + tr(L)}\frac{f_1}{f_2^2},$$

$$Y_2 = \frac{1}{-u' + tr(L)}\frac{1}{f_1}, \quad \mathcal{L} = \frac{1}{-u' + tr(L)}.$$

for $i \in \{1, 2\}$.

Remark 3.3.3. *Variations of this change of variables has been used successfully by a variety of authors (for instance, [11], [25], [47], [43]).*

Lemma 3.3.4. *The solutions constructed in (3.3.1) satisfy the smoothness conditions (3.2.1). Hence the variables $(\mathcal{L}, X_i, Y_i)$ for $i \in \{1, 2\}$ satisfy:*

$$\lim_{r\to 0}\frac{\mathcal{L}(r)}{r} = 1, \quad \lim_{r\to 0} X_1(r) = 1, \quad \lim_{r\to 0} X_2(r) = 0, \quad \lim_{r\to 0} Y_1(r) = 0, \qquad \lim_{r\to 0} Y_2(r) = \frac{2}{k}.$$

Proof. Notice that $\mathcal{L} = \frac{1}{\xi}$, which is unbounded at $r = 0$. Hence, $\mathcal{L}(0) = 0$. Then,

$$\lim_{r\to 0} \frac{\mathcal{L}}{r} = \lim_{r\to 0} \mathcal{L}' = \lim_{r\to 0} \frac{L_1^2 + 2L_2^2}{L_1^2 + 2L_2^2} = 1$$

by L'Hôpital's rule. Next,

$$\lim_{r\to 0} X_1 = \lim_{r\to 0} \frac{\mathcal{L} f_1'}{f_1} = \lim_{r\to 0} \frac{\mathcal{L}' f_1'}{f_1'} = 1.$$

The X_2 limit follows from $f_2'(0) = 0$. Similarly, the Y_1 limit follows from $f_1(0) = 0$. Finally,

$$\lim_{r\to 0} Y_2 = \lim_{r\to 0} \frac{\mathcal{L}}{f_1} = \lim_{r\to 0} \frac{\mathcal{L}'}{f_1'} = \frac{2}{k}.$$

□

By defining a function, $r(s)$ so that $\frac{dr(s)}{ds} = \mathcal{L}(r(s))$, $r(0) = T$ and $\lim_{s\to-\infty} r(s) = 0$, we arrive at the system

$$\frac{d\mathcal{L}}{ds} = \mathcal{L}(X_1^2 + 2X_2^2) \tag{3.13}$$
$$\frac{dX_1}{ds} = \frac{Y_1^2}{2} - X_1 + X_1(X_1^2 + 2X_2^2) \tag{3.14}$$
$$\frac{dX_2}{ds} = \frac{Y_2^2}{2} - \frac{(Y_1 - Y_2)^2}{2} - X_2 + X_2(X_1^2 + 2X_2^2) \tag{3.15}$$
$$\frac{dY_1}{ds} = Y_1(X_1 - 2X_2 + X_1^2 + 2X_2^2) \tag{3.16}$$
$$\frac{dY_2}{ds} = Y_2(-X_1 + X_1^2 + 2X_2^2) \tag{3.17}$$

by the chain rule. Notice that the last four equations do not depend on $\mathcal{L}$ which simplifies the analysis. Note that the "time" (r), metric, and soliton potential can be recovered from (3.13)-(3.17) via:

$$r(s) = r(s_0) + \int_{s_0}^{s} \mathcal{L}(\tau)d\tau,\ f_1 = \frac{\mathcal{L}}{Y_2},\ f_2 = \frac{\mathcal{L}}{\sqrt{Y_1 Y_2}},\ u' = \frac{X_1 + 2X_2 - 1}{\mathcal{L}}.$$

For the rest of the thesis, we use the following fact from [43] that a solution to the soliton equations (3.6)-(3.10) can be extended provided $\xi > 0$ without further reference.

3.4 Existence of complete solutions

By Theorem 3.3.1, there exists a family of solutions that depend continuously on (α, β). In this section, we study how varying (α, β) can produce complete steady solitons. To that end, consider the compact, closed set,

$$U = \{(\alpha, \beta) \mid \alpha^2 + \beta^2 = 1,\ \alpha \geq 0,\ \beta \geq 0\}.$$

Note that if $\alpha < 0$, then

$$\alpha = -u''(0) = \lim_{s\to 0} -\frac{u'(r)}{r} < 0$$

implies $u' > 0$. However, [11, Proposition 2.3] implies that u is a strictly decreasing function if the solution is complete. Hence, we can ignore the case $\alpha < 0$. In addition, we require $\beta > 0$ to satisfy the smoothness conditions (3.2.1) at $r = 0$. Thus we can restrict our attention to solutions to the soliton equations (3.3)-(3.5) parameterised by (α, β) in U.

This family of $U(2)$-invariant steady solitons was discovered independently by M. Stolarski in [42] and M. Wink in [47]. A. Appleton in [1] was then able to show that there exists a non-collapsed soliton amongst these previously constructed solitons. We follow the approach taken by T. Buttsworth in [43] by first studying the solutions $(\alpha, \beta) \in U$ by considering a neighbourhood of $(\alpha, \beta) = (1, 0)$.

Proposition 3.4.1 (Proposition 4, [43])**.** *A trajectory of (3.13)-(3.17) starting at $\phi(1, 0)$ exists for all large s and the functions $\mathcal{L}, X_1, X_2, Y_1, Y_2$ all have real valued limits as s tends to infinity. Precisely,*

$$\lim_{s\to\infty} \mathcal{L}(s) > 0 \quad \lim_{s\to\infty} X_1 = \lim_{s\to\infty} X_2 = \lim_{s\to\infty} Y_1 = 0, \quad \lim_{s\to\infty} Y_2 > 0.$$

Note that $(1, 0)$ does not give a steady soliton metric due to the uniform vanishing of Y_1. However, we can find complete metrics in some small neighbourhood in U about $(1, 0)$. To that end, notice that the the point $(0, 0, 0, a)$ is a critical point of the last four equations (3.13)-(3.17). T. Buttsworth uses this observation to obtain more solutions that satisfy the hypothesis of Proposition 3.4.1 via center manifold theory. These solutions are termed "Biaxialsolitons", denoted $U_{\text{BiaxialSolitons}}$ and it is then shown that $U_{\text{BiaxialSolitons}}$ is open in U [43, Proposition 5]. We will see that these solutions are *collapsed*.

The key property of the equations (3.3)-(3.5), is that the nature of the corresponding solutions change as we move towards $(0, 1)$ in U if $k \geq 3$. Indeed, there is a neighbourhood of $(\alpha, \beta) = (0, 1)$ in which the corresponding solutions of (3.13) are incomplete [43, Proposition 6]. This was first demonstrated by Appleton in [1] and used to construct non-collapsed solitons.

Remark 3.4.2. *T. Buttsworth in [43] shows that if $k = 1$, then the metric (2.3.11) is Ricci-flat and complete and coincides with the* Taub-Bolt metric. *Moreover, if $k = 2$ and $(\alpha, \beta) = (1, 0)$, we get another complete Ricci-flat metric which can be identified as the Kahler Eguchi-Hanson metric.*

Theorem 3.4.3 (Theorem 6, [43])**.** *If $k \geq 3$, then there exists a non-empty closed and connected set, $U_0 \subset U$ such that:*

- *U_0 does not contain $(1, 0)$ or $(0, 1)$;*
- *for any $(\alpha, \beta) \in U_0$, the corresponding solution of the soliton equations (3.13)-(3.17) starting at $\phi(\alpha, \beta)$ exists for all $s \in \mathbb{R}$;*

- *the limits as $s \to \infty$ of $\mathcal{L}, X_1, X_2, Y_1, Y_2$ all exist and are equal to zero except $\mathcal{L}$ which converges to a positive number.*

In Section 3.5 we show that solutions that satisfy the hypothesis of (3.4.3) are non-collapsed. Moreover, an equivalent interpretation of Conjecture 1.1.1 is that U_0 consists of a single point. The key property of the solutions in (3.4.3) we exploit is that they are convergent to the origin in $\mathbb{R}^4$.

3.5 Asymptotic properties

In this section we examine the asymptotic properties of solutions to (3.3) that will be relevant to our investigation. Furthermore, we also show that Appleton's soliton is *non-collapsed*.

As a first step, we examine the quantity

$$Q := \frac{Y_1}{Y_2}. \tag{3.18}$$

As remarked in (1.2), the quantity Q naturally arises in determining the completeness of solutions constructed in (3.3.1). Moreover, via the identification $Q = \frac{f_1^2}{f_2^2}$ in the r time parameterisation, we will see that Q also determines whether solutions are collapsed or non-collapsed. Indeed, Q is perhaps the most naturally occurring scale invariant quantity in this setting and variations of this approach have been taken by the authors of ([1], [47]), [4]).

From the soliton equations (3.13)-(3.17), it follows that,

$$Q' = 2Q(X_1 - X_2) \tag{3.19}$$

and hence,

$$Q'' = 2\left(X_1 - X_2 + \frac{X_1^2 + 2X_2^2}{2} + \frac{1}{2}\right)Q' + 2Y_2^2 Q(Q^2 - Q) \tag{3.20}$$

which gives us the following key lemma, which characterises the behaviour of Q,

Lemma 3.5.1. *Let $s_0 > 0$ and $(\mathcal{L}, X_i, Y_i) : [0, s_0) \to \mathbb{R}^3$ be a solution of the equations 3.13. Suppose $s \in [0, s_0)$ is a local extrema of Q. If $Q(s) > 1$ ($0 \leq Q(s) < 1$), then Q has a strict local minimum (maximum) at s.*

Proof. This follows from the evolution equation (3.20). □

A. Appleton in [1] uses this lemma as a starting point for the analysis of his solutions.

Remark 3.5.2. *Since $Q(0) = 0$, it follows that for some $s_0 > 0$, if $Q(s_0) > 1$ then $Q'(s) > 0$ for all $s > s_0$.*

We now consider the monotonicity properties of f_1, f_2.

Lemma 3.5.3. *Let $s_0 > 0$ and $(u, f_1, f_2) : I \to \mathbb{R}^3$ be a solution to the soliton equations (3.3)-(3.5). Then f_1 is strictly increasing on $[0, s_0)$ and f_2 is strictly increasing on any interval $0 \in I \subset [0, s_0)$ where $Q < 2$. Moreover, f_2' changes sign at most once on the interval $[0, s_0)$.*

Proof. Note that equation (3.3) implies that $f_1'' = \frac{f_1^3}{2f_2^4}$ when $f_1' = 0$. The boundary conditions (3.2.3), imply that $f_1'(0) > 0$ and hence f_1 is strictly increasing.

The evolution equation of f_2 in (3.3), we get that $f_2'' = \frac{1}{2}\frac{2-Q}{f_2}$ when $f_2' = 0$. Applying L'Hôpital's rule at $s = 0$ implies $f_2''(0) > 0$ and thus $f_2'(0) > 0$ from Lemma 3.2.3, implies that f_2 is strictly increasing on any interval $I = [0, s]$, $s > 0$ where $Q < 2$. Hence f_2' can change its sign only when $Q \geq 2$. Since Q is strictly increasing when $Q > 1$, and $f_2'' = \frac{1}{2}\frac{2-Q}{f_2}$ when $f_2' = 0$, it follows that f_2 changes its sign at most once. □

The following Lemma from [1] then establishes the behaviour of complete solutions.

Lemma 3.5.4. *For any $C > 0$, there exists a $\gamma_0 > 0$ such that a solution $(u, f_1, f_2) : [0, \infty) \to \mathbb{R}^3$ to the equations (3.3)-(3.5), with $\gamma < -\gamma_0$ satisfies $Q \leq C$ everywhere.*

Remark 3.5.5. *For complete solutions, Lemma's 3.5.4 and 3.5.3 in conjunction with the fact $\lim_{s\to\infty} \mathcal{L}(s) \to C > 0$ as $\xi(r) > 0$ for complete steady solitons [11, Proposition 2.4], imply that Y_1 and Y_2 are both bounded from above.*

By adapting *only* the methods of A. Appleton, we believe that it is possible to show that $Q \leq 1$ for all complete solitons in our case. However, the key insight of T. Buttsworth is the applicability of center manifold techniques to this problem and the consequential simplification of the problem. Indeed, using center manifold techniques, T. Buttsworth was able to show that $Q \leq 1$ for complete solitons in [1]. In our case, T. Buttsworth found the following asymptotic properties.

Proposition 3.5.6. *If $(\alpha, \beta) \in U_0$, then*

$$f_i(r) \sim \sqrt{r},\ f_i' \sim \frac{1}{\sqrt{r}},\ f_i'' \sim \frac{1}{\sqrt{r^3}}$$

for $i \in \{1, 2\}$. However, if (α, β) is such that $\lim_{\to\infty} Y_2 > 0$, then,

$$\lim_{r\to\infty} f_1(r) > 0,\ f_2(r) \sim \sqrt{r}$$

where the notation $A \sim B$ denotes the existence of constants, $C, C' > 0$ such that $C'B \leq A \leq CB$.

It turns out that the more center manifold heavy approach proposed by T. Buttsworth in [43] simplifies the analysis in some sense. However, in order to apply these techniques, we need to understand the asymptotic properties of our solution. The key result in this direction is the following proposition from T. Buttsworth in [43].

Proposition 3.5.7 (Proposition 7, [43])**.** *For each $\delta' > 0$, there is an open and connected neighbourhood $\tilde{U} \subset U$ of U_0 and a large enough T_0 such that:*

- *if a solution* $(\xi, L_1, L_2, R_1, R_2)$ *to the soliton equations (3.13)-(3.17), corresponding to* $(\alpha, \beta) \in \tilde{U}$ *exists for all* $r \geq 0$*, then* (X_1, X_2, Y_1, Y_2) *lies in* $B_{\frac{\delta}{2}}(0)$ *for all* $s \geq T_0$*; and,*

- *for any* $(\alpha, \beta) \in \tilde{U}$*, the corresponding solution exists until time* T_0 *and* (X_1, X_2, Y_1, Y_2) *at* $s = T_0$ *lies in* $B'_{\frac{3}{4}\delta}$*.*

We note that the proof of (3.5.7) relies on the conservation laws discussed at the beginning of this section. Moreover (3.5.7) tells us that complete solutions in some neighbourhood of the solutions of interest, U_0, get arbitrarily close to the origin. It is this crucial property that allows us to pass to trajectories on the center manifold which we describe in the next section. We conclude this section by showing that the non-collapsed solitons in U_0 (as in 3.4.3) are indeed non-collapsed.

Lemma 3.5.8. *The solutions to the soliton equations (3.3)-(3.5), satisfy* $R \sim \frac{1}{r}$ *where* R *is the scalar curvature.*

Proof. Lemma 3.5.6 implies that f_i' decays like $\frac{1}{\sqrt{r}}$ while the functions f_1, f_2 grow like $\sqrt{r}$. Hence, $\frac{f_i'}{f_i} \sim \frac{1}{r}$ for $i \in \{1, 2\}$ and the conclusion follows. □

With this estimate on the curvature decay, we can show that the claimed non-collapsed solution is actually non-collapsed.

Proposition 3.5.9. *A solution to the soliton equations (3.3)-(3.5) corresponding to some* $(\alpha, \beta) \in U_0$ *is non-collapsed.*

For the following proof, we fix the notation, s to refer to the radial coordinate in $SU(2) \times_{U(1)} \mathbb{R}^2$ and r to be the radius of the ball.

Proof. Let $p \in M_0$ be such that p lies on a geodesic, $\gamma : I \to M$, that intersects the $SU(2)$ orbits perpendicularly such that $p := [s_0, x_0] = \gamma(t_0)$. Then, there exists an open neighbourhood, U of p such that γ is distance minimising and $q := [s_1, x_1] = \gamma(t_1) \in U$ where $s_1 - s_0 > 0$. Hence, we can assume that $d(p, q) < \epsilon$ for $\epsilon > 0$ where d is the distance function (2.2) and that $q \in B(p, \epsilon)$.

Next, suppose that $|Rm| \leq \frac{1}{r^2}$. Consider $r = s_1 - s_0 > 0$ where s_1, s_0 are the radial coordinates of the points $p, q \in M$ as above. Then, as $\mathrm{R} \leq n^2|Rm|$ and $\mathrm{R} \sim \frac{1}{s}$, we observe that if the radius, r satisfies

$$\frac{1}{s_0} \leq \frac{1}{r^2} = \frac{1}{(s_0 - s_1)^2}$$

(up to constants) such that $\mathrm{Vol}B(p, r) \geq \kappa r^4$ for $\kappa > 0$ as in (2.1.20), then the solution is non-collapsed. Hence, $s_1 < s_0 + \sqrt{s_0}$ (i.e. $r = s_1 - s_0 < \sqrt{s_0}$). Thus, by setting $s = \frac{s_1 + s_0}{2}$ and $r' = \frac{s_1 - s_0}{2}$, we can assume that the maximal allowable radius at a point $p \in M$ with radial coordinate, s satisfies

$$r = \mathcal{O}(\sqrt{s})$$

Recall from the proof of (3.5.8), that the functions, f_i grow like $\sqrt{s}$. Hence, we can bound the volume form, dV via

$$dV \geq C_1 \sqrt{s}^3 ds \wedge \omega^1 \wedge \omega^2 \wedge \omega^3$$

for some $C_1 > 0$. By passing to spherical coordinates (ψ, θ, φ) on S^3, we obtain

$$VolB(p,r) \geq \int_{s_0}^{s_1} \int_0^{2\pi} \int_0^{\pi} \int_0^{\pi} \sqrt{s}^3 (\sin^2 \varphi \ \sin\theta) \ d\psi \wedge d\theta \wedge d\varphi \wedge ds \geq C_2 s^{5/2}$$

for some constant $C_2 > 0$. Finally, as $r = \mathcal{O}(\sqrt{s})$,

$$\frac{VolB(p,r)}{r^4} \geq \frac{C_2 s^{\frac{5}{2}}}{C' s^2} = C'' \sqrt{s}$$

for constants $C', C'' > 0$ and the conclusion follows.

□

Remark 3.5.10. *With the same method, one can show that if $f_1 \sim 1$, and $f_2 \sim \sqrt{s}$, then the corresponding solution is collapsed.*

Chapter 4

Center manifold

The key motivation for this section is that the last four equations of the non-collapsed solution in (3.4.3) converge to the origin in $\mathbb{R}^4$ *and* the origin is a critical point of the equations (3.13)-(3.16). Furthermore, the trajectories in some open of U_0 get "close" to the origin. Thus, it turns out that trajectories on a time-invariant manifold, called the "center manifold" are *equivalent* in some sense to our trajectories of interest. The goal of this section will be to establish and study this equivalence in order to study our solutions of interest.

Center manifold theory is a dynamical systems technique, used to study the dynamics of a system about non-hyperbolic equilibrium points. In our case, the equations (3.13)-(3.16), have an equilibrium point at the origin and thus we can use center manifold techniques to analyse its stability. To fix notation, we write (3.13)-(3.16) as

$$\dot{z} = f(z) \tag{4.1}$$

where $f : \mathbb{R}^4 \to \mathbb{R}^4$ is the corresponding smooth function. Then $f(0_{\mathbb{R}^4}) = 0$. Denote the trajectory of (4.1) with initial value, $z(0) = z^*$ as

$$t \mapsto z(t) =: \phi(t, z^*). \tag{4.2}$$

Let $A = Df(0)$ be the Jacobian matrix of f at the origin. Then we can write (4.1) as

$$\dot{z} = Az + g(z) \tag{4.3}$$

where $g(0) = 0$ and $Dg(0) = 0$. At these non-hyperbolic points, one can decompose $\mathbb{R}^4$ into at most three subspaces, V^s, V^c, V^u called *stable*, *center* and *unstable* subspaces defined by the span of the negative, zero and positive eigenvalues of A respectively. That is,

$$\mathbb{R}^4 = V^s \oplus V^c \oplus V^u$$

with continuous projections from $\mathbb{R}^4 \to V^i$ denoted by π_s, π_c, π_u respectively and inclusions $i : V^i \hookrightarrow \mathbb{R}^n$. In the case where the center subspace, V^c vanishes, the well-known *Hartman-Grobman theorem* implies that the linearised flow, $\dot{z} = Az$ and the non-linear flow (4.1) are equivalent in some neighbourhood about the equilibrium point. However, when V^c is non-trivial, we apply the *center manifold theorem* given in [7]. This theorem tells us that the interesting dynamics take place on an invariant manifold, $\mathcal{M}$, tangent to the center subspace

V^c. One advantage of this identification is its *dimension reduction property*. Instead of studying the flow on the entire space $\mathbb{R}^4$, we can restrict our analysis to the *center manifold* of the same dimension as V^c. The precise statement of this theorem from [7] is as follows.

Theorem 4.0.1. *let $f : \mathbb{R}^n \to \mathbb{R}^n$ be a vector field in $\mathbb{C}^{k+1}$ ($k \geq 1$) with $f(0) = 0$. Consider the matrix $A = Df(0)$ and let V^s, V^c, V^u denote the stable, unstable and center subspaces respectively. Then there exists $\delta > 0$, and a local center manifold, $\mathcal{M}$ with the following properties.*

1. *There exists a $\mathbb{C}^k$ function $\varphi : V^c \to \mathbb{R}^n$ with $\pi_c\varphi(z_c) = z_c$ such that*

$$\mathcal{M} = \{\varphi(z_c) \mid z_c \in V^c,\ |z_c| < \delta\}.$$

2. *If $x \in \mathcal{M}$, then $\phi(t, z) \in \mathcal{M}$ for $|t| < \epsilon$, $\epsilon > 0$.*

3. *The center manifold, $\mathcal{M}$ is tangent to V^c at the origin.*

4. *If $\phi(t, z) \in B_{\delta'}(0)$ for some δ' and all $t \in \mathbb{R}$, then $\phi(t, z) \in \mathcal{M}$ for all $t \geq 0$. In other words, globally bounded orbits in some suitably small neighbourhood of the origin are contained inside $\mathcal{M}$.*

5. *Given any trajectory $x(t) \to 0$ as $t \to \infty$, there exists an $\eta > 0$ and a trajectory $t \mapsto y(t) \in \mathcal{M}$ on the center manifold such that*

$$e^{\eta t}|x(t) - y(t)| \to 0, \quad \text{as } t \to \infty$$

where $\eta \in (0, \beta)$ where β is the spectral gap of A in (4.3).

A key property that can be derived from the proof of Theorem 4.0.1 is that one can establish an equivalence between trajectories on the center manifold and trajectories from (4.1) that lie in some small ball about the origin. For instance, [43, Theorem 6] defines a map, $h : \mathbb{R}^5 \to \mathcal{M}$ and shows that for some trajectory $\phi(t, z) \in B_\delta(0)$ for $t \geq 0$, the corresponding trajectory on the center manifold, $\phi(t, h(z))$ converges exponentially quickly to the original trajectory as $t \to \infty$. The proof of this fact will follow by from the proof of part 5 of Theorem 4.0.1.

Remark 4.0.2. *A good introductory reference for the standard theory of center manifolds can be found in [46].*

As the proof of this theorem can be found in [7], we will not repeat it. Instead, we detail the parts relevant to our purposes. We will restrict our attention to the last four equations of the (3.13) system since these determine the behaviour of the first equation and are independent of the first equation. Hence,

Lemma 4.0.3. *The system (3.13)-(3.16) written as $\dot{z} = f(z)$ satisfies $f(0) = 0$. In particular, if $A := Df(0)$, we find that*

$$A = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Hence if we consider the standard orthonormal basis, $\{e_i\}_{i=1}^4$ of $\mathbb{R}^4$, we see that:

$$V^s = span\{e_1, e_2\}, \quad V^c = span\{e_3, e_4\}.$$

By considering the Taylor expansion about zero, it follows that higher order terms satisfy

$$g(x_1, x_2, y_1, y_2) = \begin{pmatrix} \frac{y_1^2}{2} + x_1(x_1^2 + 2x_2^2) \\ \frac{y_2^2 - (y_1 - y_2)^2}{2} + x_2(x_1^2 + 2x_2^2) \\ y_1(x_1 - 2x_2 + x_1^2 + 2x_2^2) \\ y_2(-x_1 + x_1^2 + 2x_2^2) \end{pmatrix}$$

for $g \in \mathbb{R}[x_1, x_2, y_1, y_2]$ and hence (3.13)-(3.16) becomes

$$\dot{z} = Az + g(z)$$

Proof. By considering the Taylor expansion correct to third order, we obtain the desired matrices. Then, the spaces V^s and V^c follow by definition. □

We also need the following lemma concerning the $\mathbb{C}^1$ norm of g. Let $\rho : \mathbb{R} \to [0, 1]$ be a smooth, even function with compact support such that

$$\rho(\xi) = \begin{cases} 1 & if |\xi| \leq 1 \\ 0 & if |\xi| \geq 2. \end{cases}$$

Thus, for $\epsilon > 0$, we define the truncated function

$$g_\epsilon(x) = \rho\left(\frac{|x|}{\epsilon}\right) g(x) \tag{4.4}$$

Then,

Lemma 4.0.4. *$\|g\|_{C^1}$ is uniformly bounded with compact support and g_ϵ coincides with g for $|x| \leq \epsilon$. In particular, $\|g_\epsilon\| \leq \mathcal{O}(1)\epsilon$.*

Proof. The proof can be found in [7, Section A.3.1] □

Thus we can construct the center manifold with these assumptions by using the fact that g coincides with g_ϵ for $|x| \leq \epsilon$.

4.1 Center manifold construction

There are two approaches when constructing the center manifold. A first approximation can be found by considering Theorem 4.0.1, (3), which states that the manifold $\mathcal{M}$ is tangent to V^c at the origin. Using this property, we can define two C^3 functions, $C_i : \mathbb{R}^2 \to \mathbb{R}$ for $i \in \{1, 2\}$ parameterising the center manifold such that $x_i = C_i(y_1, y_2)$. The method, as detailed in [46], is to use a n-th order Taylor series to obtain a general form of the maps C_i up to n-th order and then substitute it into the expression

$$x_i'(C_1, C_2, y_1, y_2) = \nabla C_i(y_1, y_2) \cdot y'(C_1, C_2, y_1, y_2) \tag{4.5}$$

which follows from this tangent condition (part 3, 4.0.1). In general, the order considered is the least such that one can extract useful properties. In [43], T. Buttsworth shows that only a single C^3 map is required to describe the center manifold. For notational convince however, we reformulate this as two functions as follows.

Lemma 4.1.1. *There is are two C^3 functions $C_1, C_2 : \mathbb{R}^3 \to \mathbb{R}$ and $\epsilon > 0$ such that the two dimensional submanifold $\mathcal{M}$ of $\mathbb{R}^4$ given by $x_1 = C_1(y_1, y_2)$, $x_2 = C_2(y_1, y_2)$ is invariant under the evolution of (3.13) for trajectories lying in $B_\epsilon(0)$, referred to as the center manifold. Explicitly,*

$$\mathcal{M} = \{(C_1, C_2, y_1, y_2) \in \mathbb{R}^4 \mid y_1, y_2 \in \mathbb{R}\}$$

where $C_i = x_i$. Furthermore, the symmetries of C, imply that if $y_1 = y_2$, then $x_1 = x_2$. In other words, $C_1 = C_2$.

To make Lemma 4.1.1 more concrete, one can consider a second order approximation given below.

Lemma 4.1.2. *There are C^3 functions, $C_1, C_2 : \mathbb{R}^2 \to [0, \infty)$ correct up to second order such that $x_1 = C_1(y_1, y_2)$ and $x_2 = C_2(y_1, y_2)$ given by,*

$$x_1 = C_1(y_1, y_2) = \frac{y_1^2}{2}, \tag{4.6}$$

$$x_2 = C_2(y_1, y_2) = y_1 y_2 - \frac{y_1^2}{2}. \tag{4.7}$$

Proof. The proof follows by considering a general second order Taylor series for the functions C_1, C_2 and substituting them into Equation 4.5. □

Lemma 4.1.2 allows us to examine the dynamics on the center manifold numerically correct to second order. Recall that trajectories on the center manifold satisfy

$$\dot{\mathbf{y}} = A\mathbf{y} + g(C(\mathbf{y}), \mathbf{y})) \tag{4.8}$$

and thus we can numerically simulate the trajectories, (4.8) as in Figure 4.1.
This shows some of the essential features of the center manifold. For a start, the dynamics are *slow* due to the proximity of the equilibrium point. Furthermore, there are three distinct regions:

Proposition 4.1.3. *The center manifold, $\mathcal{M}$, has three distinct regions with special properties. Precisely, if $(\mathcal{L}, X_1, X_2, Y_1, Y_2) : (-\infty, T_0) \to \mathbb{R}^5$ for $T_0 > 0$ is a solution to Theorem 3.3.1 that lies on $\mathcal{M}$, then:*

- *if $Y_1 > Y_2$ then the solution is incomplete,*
- *if $Y_2 > Y_1$ then the solution is complete and collapsed,*
- *if $Y_1 = Y_2$ then the solution is complete and non-collapsed,*

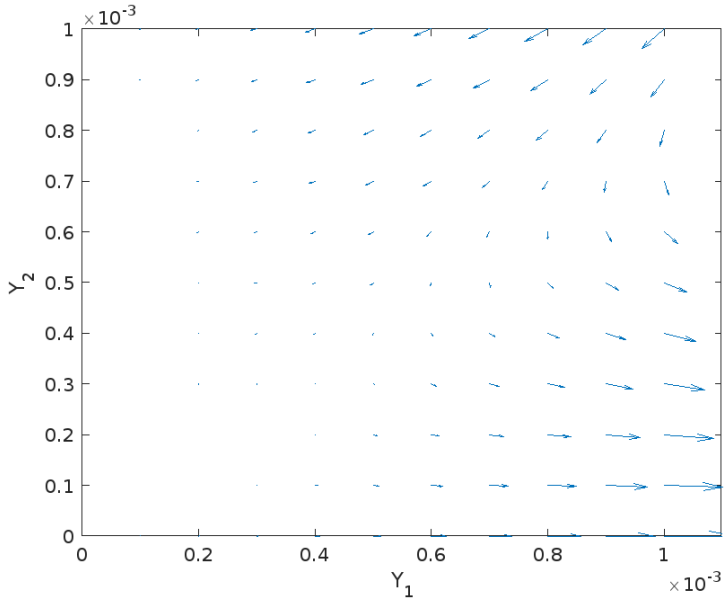


Figure 4.1: Vector field plot correct to second order about the origin.

Note that if $Y_1 > Y_2$, then the work of T. Buttsworth in [43, Proposition 7] implies that the solution is incomplete. Moreover, Buttsworth in [43] showed the converse holds. Thus if $Y_1 \leq Y_2$, the solutions are complete. Hence it suffices to show that if $Y_1 = Y_2$, the limit of *just* these two is zero and furthermore, if $Y_1 < Y_2$, then $Y_1 \to 0$ as $Y_2 \to a > 0$ for some constant a.

Proof. Let C_1, C_2 be two C^3 functions of y_1, y_2 such that $x_1 = C_1(y_1, y_2)$, $x_2 = C_2(y_1, y_2)$ as described in Lemma 4.1.1. Then, by Equation 3.13,

$$\begin{aligned} C_1(C_1^2 + 2C_2^2 - 1) + \frac{Y_1^2}{2} &= \frac{dX_1}{ds} \\ &= \frac{dC_1}{ds} \\ &= \frac{dC_1}{dy_1} y_1(C_1 - 2C_2 + C_1^2 + 2C_2^2) + \frac{dC_1}{dy_2} y_2(-C_1 + C_1^2 + 2C_2^2). \end{aligned}$$

Next, the proof of [43, Lemma 3] implies that if $y_1^2 + y_2^2$ is sufficiently small, then the quantity $-C_i + C_i^2 + 2C_j^2$ is non-positive and zero if and only if $Y_1 = 0$. Furthermore, Equation 3.13 implies that

$$\frac{dY_1}{ds} = Y_1(C_1 - 2C_2 + C_1^2 + 2C_2^2), \quad \frac{dY_2}{ds} = Y_2(-C_1 + C_1^2 + 2C_2^2)$$

Hence if $Y_1 = Y_2$, then $C_1 = C_2$ and thus Y_1, Y_2 converge to zero. However, if $Y_1 < Y_2$, we can consider the second order approximations of C_i to conclude that Y_1 decays faster than Y_2 and hence $Y_1 \to 0$ while $Y_2 \to C$. Explicitly,

$$\begin{aligned}\frac{dY_1}{ds} &= Y_1(C_1 - 2C_2 + C_1^2 + 2C_2^2) \\ &< Y_1(-Y_1^2 + C_1^2 + 2C_2^2) \\ &< Y_2(-\frac{Y_1^2}{2} + C_1^2 + 2C_2^2) = \frac{dY_2}{ds}\end{aligned}$$

For completeness, recall that $\frac{1}{\mathcal{L}} = \xi$. Thus [11, Proposition 2.4] implies that $\lim_{s\to\infty} \xi(s) = \sqrt{-C}$ for $C < 0$ and that $(u')^2 \leq R(0) < 2\alpha$. Hence, $\lim_{s\to\infty} \mathcal{L} < \infty$. □

4.2 The Gamma map

We now come to constructing the center manifold embedding in $\mathbb{R}^4$. We will adapt the proof found in [7] such that the resulting equivalence can be numerically simulated. We will then analyse the error that arises from these approximations. We note that the algorithm proposed herein works in the setting of a general Picard iteration.

As a first step, define the Banach space,

$$\mathcal{Y} = \left\{ y \in C^0(\mathbb{R}, \mathbb{R}^n) \;\middle|\; \|y(\cdot)\| := \sup_{t\in\mathbb{R}} e^{-|t|/2}|y(t)| < \infty \right\}$$

Remark 4.2.1. *The quantity $\frac{1}{2}$ is arbitrary. One could chose any $\eta \in (0,1)$ (i.e. in the spectral gap of A).*

and $\Gamma : V^c \times \mathcal{Y} \to \mathcal{Y}$ given by,

$$\Gamma(x_c, y)(t) = e^{At}x_c + \int_0^t e^{A(t-\tau)}\pi_c g(y(\tau))d\tau + \int_{-\infty}^t e^{A(t-\tau)}\pi_s g(y(\tau))d\tau.$$

We will use the contraction mapping principle to show that $\Gamma : V^c \times \mathcal{Y} \to \mathcal{Y}$ has a unique fixed point for each $x_c \in V^c$ denoted $y^{x_c} := y(0)$, where $y(\cdot) \in \mathcal{Y}$. Then,

$$\mathcal{M} = \{x_c \in V^c \mid y^{x_c}(0)\} \tag{4.9}$$

by [7, Section A.3.2]. As $x_c \in \text{span}\{e_3, e_4\}$, this implies that $e^{At}x_c = i_c(x_c)$ since A is diagonal and $e^0 = 1$. Hence,

$$\Gamma(x_c, y)(t) = i_c(x_c) + \int_0^t \pi_c g(y(\tau))d\tau + \int_{-\infty}^t e^{A(t-\tau)}\pi_s g(y(\tau))d\tau.$$

To show that the map $y \mapsto \Gamma(x_c, y)$ is a strict contraction, we suppose that $\|g\|_{\mathbb{C}^1} \leq \frac{1}{2}C$ for $C > 0$ via Lemma 4.0.4. Then,

$$\|\Gamma(x_c, y_1) - \Gamma(x_c, y_2)\| \leq \frac{1}{2}\|y_1 - y_2\|$$

and the contraction mapping theorem, the map, Γ has a fixed point, y^{x_c} as required.

The issue from an analytic standpoint, is that an explicit closed form expression for the iterated map Γ is complex. Hence, we use numerical methods to evaluate Γ so that we can approximate our center manifold numerically.

To deal with the map Γ taking values, $t \in (-\infty, \infty)$, we suppose our approximation, Γ^* is supported on $t \in [-R, R]$ for $R > 0$. We are able to achieve reasonable accuracy with this method due to the properties of the Banach space these functions belong to. We first present the MATLAB based pseudo-code for our Γ iteration. To fix notation, let $N \in \mathbb{N}$ be the number of iterations to perform, $R \in \mathbb{R}_{>0}$ such that $|t| \leq R$ and $K \in \mathbb{N}$ the number of integration points. Then we can iterate Γ as in Algorithm 1.

Algorithm 1 Simple Γ iteration routine.

$t \leftarrow$ linspace(-R, R, 2K-1)
$y \leftarrow$ zeros(4, 2K-1)
$i \leftarrow 1$
while $i \leq N$ **do**
    $y \leftarrow \Gamma(x_c, y)(t)$
end while

Then Γ is calculated numerically below in Algorithm 2.

Algorithm 2 Γ

$g_\epsilon[j] \leftarrow g(y)[j]$
$I_1 =$ Array(2, K)
$I_2 =$ Array(2, K)
$t_0 = t[t == 0]$
$i \leftarrow 0$
while $i \leq K$ **do**
    if $i \leq t_0$ **then**
        $I_1[:, i] \leftarrow \text{trapz}(t(i : t_0), g_\epsilon(3 : 4, i : t_0), 2)$
    else
        $I_1[:, i] \leftarrow \text{trapz}(t(t_0 : i), g_\epsilon(3 : 4, t_0 : i), 2)$
    end if
end while
$i \leftarrow 0$
while $i \leq K$ **do**
    $z \leftarrow \text{zeros}(2, i)$
    $j \leftarrow 0$
    while $j \leq i$ **do** $z(:, j) \leftarrow e^{t(j)-t(i)} g_\epsilon(1 : 2, j)$
    end while
    $I_2(:, i) \leftarrow \text{trapz}(t(1 : i), z(:, 1 : i), 2)$
end while
$y \leftarrow x_c + [I_2; I_1]$

Note that we can ensure that our numerical function, y, is in Y_η by assuming that y is supported on $[-R, R]$.

4.2.1 Error estimates

Now we give some error estimates for our approximation on the domain $|t| \leq R$. Fix $x_c \in V^c$ and $y \in Y_\eta$. Then, we are concerned with the quantity,

$$\|\Gamma(x_c, y) - \Gamma^*(x_c, y)\| = \sup_{t\in\mathbb{R}} e^{\frac{-|t|}{2}} |\Gamma(x_c, y) - \Gamma^*(x_c, y)|$$

where Γ is the actual map and Γ^* the approximation supported on $[-R, R]$. To be explicit,

$$\Gamma^*(x_c, y)(t) = i(x_c) + \int_0^t \pi_c g(y(\tau))d\tau + \int_{-R}^t e^{A(t-\tau)}\pi_s g(y(\tau))d\tau \tag{4.10}$$

where $|t| < R$.

Lemma 4.2.2. *Let $y, y^* \in \mathcal{Y}$ distinct and $|t| \leq R$. Then,*

$$\sup_{|t|<R} e^{\frac{-|t|}{2}} |\Gamma(x_c, y)(t) - \Gamma^*(x_c, y^*)(t)| \leq 2\epsilon R + 5\epsilon$$

for some $\epsilon > 0$.

Proof. First note that $\|\Gamma(x_c, y)(t) - \Gamma^*(x_c, y^*)(t)\| \leq |\Gamma(x_c, y)(t) - \Gamma^*(x_c, y^*)(t)|$ since $|t| < R$. Hence,

$$\begin{aligned}
|\Gamma(x_c, y)(t) - \Gamma^*(x_c, y^*)(t)| &= \Big| \int_0^t \pi_c(g_\epsilon(y(\tau)) - g_\epsilon(y(\tau)))d\tau + \int_{-\infty}^{-R} e^{\tau+R}\pi_s g_\epsilon(y(\tau))d\tau \\
&\quad + \int_{-R}^t e^{A(t-\tau)}\pi_s(g_\epsilon(y(\tau)) - g_\epsilon(y^*(\tau)))d\tau\Big| \\
&\leq |2\epsilon t + \epsilon + 2\epsilon(1 + e^{-R-t})| \\
&\leq 2\epsilon R + 5\epsilon
\end{aligned}$$

since we can assume g is uniformly bounded by Lemma 4.0.4. □

On the other hand,

Lemma 4.2.3. *Let $y, y^* \in Y$, distinct and $|t| \geq R$. Then,*

$$\sup_{|t|<R} e^{\frac{-|t|}{2}} |\Gamma(x_c, y)(t) - \Gamma^*(x_c, y^*)(t)| \leq \delta R e^{-\frac{R}{2}}$$

for some small $\delta > 0$.

Proof. Suppose $t > R$. Then,

$$\begin{aligned}
|\Gamma(x_c, y)(t) - \Gamma^*(x_c, y^*)(t)| \leq \Big| &\int_0^R \pi_c(g(y(\tau)) - g(y^*(\tau)))d\tau + \int_R^t \pi_c g(y(\tau))d\tau + \\
&\int_{-\infty}^{-R} e^{\tau+R} g(y(\tau))d\tau + \int_{-R}^R e^{\tau-R} g(y(\tau)) - g(y^*(\tau))d\tau + \int_R^t e^{\tau-t} g(y(\tau))d\tau\Big| \\
&\leq |2R\epsilon + \epsilon(t - R) + \epsilon(1 - e^{-2R}) + \epsilon + \epsilon(1 - e^{R-t})| \\
&= |\epsilon(t - R + 3) - \epsilon(e^{-2R} + e^{-R-t})|
\end{aligned}$$

since we can assume that g has uniformly bounded by $\epsilon > 0$ (4.0.4). Then

$$\begin{aligned} e^{-|t|/2}|C(t)| &= |e^{-|t|/2}\epsilon(t - R + 3) - \epsilon(e^{-|t|/2-2R} + e^{-|t|/2-R-t})| \\ &\leq 2\epsilon e^{\frac{-R+1}{2}} + 2\epsilon e^{-\frac{5}{2}R} \leq 4\epsilon e^{-\frac{R}{2}} \end{aligned}$$

On the other hand, suppose that $t < -R$. Then,

$$\begin{aligned} |\Gamma(x_c, y)(t) - \Gamma^*(x_c, y^*)(t)| &\leq \left| \int_0^{-R} g(y(\tau)) - g(y^*(\tau))d\tau + \int_{-R}^{t} g(y(\tau))d\tau + \int_{-\infty}^{t} e^{\tau - t} g(y(\tau))d\tau \right| \\ &\leq 2R\epsilon + \epsilon(t + R) + \epsilon. \end{aligned}$$

Hence

$$e^{-\frac{|t|}{2}}|2R\epsilon + \epsilon(t + R) + \epsilon| \leq \epsilon e^{-\frac{R}{2}}(R + 1) + 2\epsilon^{\frac{-R-2}{2}} \leq \delta R e^{-\frac{R}{2}}$$

for some $\delta > 0$. □

By taking a supremum over the regions described in Lemma's 4.2.2 and 4.2.3, the following Lemma over $\mathbb{R}$ is immediate.

Proposition 4.2.4. *For $\epsilon > 0$ corresponding to g_ϵ, and some $\delta > 0$, the estimate*

$$\|\Gamma(x_c, y) - \Gamma^*(x_c, y^*)\| \leq \delta R e^{-\frac{R}{2}} + 2\epsilon R + 5\epsilon$$

holds.

Thus, by choosing ϵ and R we can control the error with respect to the Banach space, $\mathcal{Y}$.

Remark 4.2.5. *The fact g_ϵ is uniformly bounded implies we can identify y^* (for instance) with the n-th iterate of our numerical approximation.*

4.2.2 Numerical simulation

With the algorithms (1) and (2) proposed in the previous section, we can numerically simulate our approximation, Γ^* given by (4.10) to draw the center manifold in MATLAB as follows.

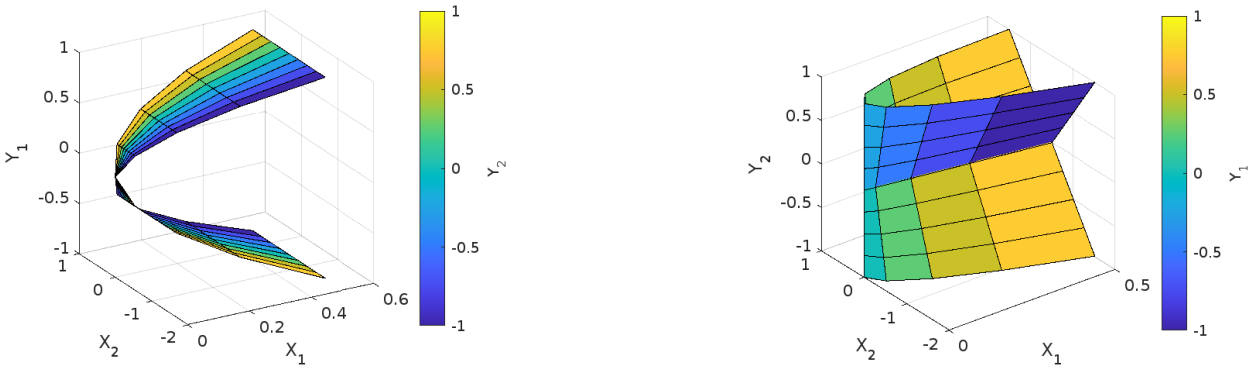


Figure 4.2: The center manifold (4.9) embedded in $\mathbb{R}^4$.

Ultimately, we want to find that there is only one choice of parameter, (α, β) such that the corresponding solution to the soliton equations (3.13)-(3.16) converges to the origin in $\mathbb{R}^4$. Figure 4.2.2 indicates that $Y_1 = 0$ coincides with $X_1 = X_2 = 0$. However, $Y_2 \geq 0$, consistent with the asymptotic properties in Section 3.5.

4.3 The Lambda map

We now construct a map $\Lambda : \mathbb{R}^4 \times W \to W$ for some Banach space, W. This will allow us to map points that are sufficiently close to the origin in $\mathbb{R}^4$ onto the center manifold. The usefulness is that the corresponding trajectories are equivalent in the sense of point 5 of Theorem 4.0.1. As in Section 4.2, we apply the contraction mapping theorem to a Picard-iteration to construct Λ.

Let $z \in \mathbb{R}^4$ such that $z \in B_\delta(0)$ for some $\delta > 0$. Moreover, suppose that the corresponding solution, $\phi(t, z)$, that coincides with z at $t = 0$ is such that $\phi(t, z) \in B_\delta(0)$ for all $t \geq 0$. We can extend this to a continuous function, $z^* : \mathbb{R} \to \mathbb{R}^4$ via

$$z^* = \begin{cases} \phi(t, z) & \text{if } t \geq 0 \\ z & \text{if } t < 0. \end{cases}$$

As a result, z^* is a *globally bounded solution* to

$$z^*(t) = Az^* + g(z^*) + \varphi(t), \quad \varphi(t) = \begin{cases} 0 & \text{if } t > 0 \\ -Az - g(z) & \text{if } t < 0. \end{cases}$$

If we consider the Banach space,

$$W = \{w \in C^0(\mathbb{R}, \mathbb{R}^4) \mid \|w(\cdot)\| := \sup_{t \in \mathbb{R}} e^{\frac{t}{2}} |w(t)| < \infty\},$$

we see that $W \subset Y$ where Y is the Banach space defined in the proof of Γ in the previous section. Furthermore, we claim that there exists $w \in W$ such that $y = z^* + w \in Y_\eta$ is a global solution to (4.0.3), contained in the center manifold, $\mathcal{M}$. Following [7], we consider the map

$$\begin{aligned} \Lambda(z, w)(t) = &\int_{-\infty}^{t} e^{A(t-\tau)} \pi_s (g(z^*(\tau) + w(\tau)) - g(z^*(\tau)) - \varphi(\tau)) d\tau \\ &+ \int_{t}^{\infty} \pi_c (\varphi(\tau) + g_\epsilon(z^*(\tau)) - g_\epsilon(z^*(\tau) + w(\tau))) d\tau \end{aligned} \tag{4.11}$$

It can be shown that the map is a strict contraction if $\|g_\epsilon\|_{\mathbb{C}^1}$ is sufficiently small. Thus, for fixed z, Λ admits a unique fixed point $w \in W$. Thus, we have a solution $y = z^* + w \in Y$ to (4.0.3) which belongs to $\mathcal{M}$. By construction, $\phi(t, z)$ coincides with z^* for all $t \geq 0$. Thus, for all $t \geq 0$,

$$|\phi(t, z) - y(t)| = |w(t)| \leq e^{\frac{-t}{2}} \|w\|.$$

This implies that we can define a map, $h : \mathbb{R}^4 \to \mathcal{M}$ such that $h : z \mapsto y^z(0)$ where w^z is the fixed point of $\Lambda(z)$ so that

$$\lim_{t \to \infty} e^{\frac{t}{2}} |\phi(t, z) - \phi(t, h(z))| = 0.$$

Due to the definition of h, we need to determine the unique fixed point w^z of Λ and evaluate it at zero. This is clearly a very similar scenario to Γ and thus much of the analysis is the same.

To this end, we approximate Λ so that it is supported on $[-R, R]$. Then let $z \in \mathbb{R}^4$ be a point, R some fixed constant in $\mathbb{R}_{>0}$, $K \in \mathbb{R}_{>0}$ the number of partitions in the trapezoidal rule and represent $w \in W$ numerically as $4 \times 2K - 1$ array. Then, we can calculate Λ as follows.

Algorithm 3 Λ

$t \leftarrow linspace(-R, R, 2K-1)$
$[s, z] \leftarrow \text{ode45}(\phi(t, z), z)$
$z \leftarrow \text{interp1}(s, z, t)$
$I_1 = \text{Array(2, K)}$
$I_2 = \text{Array(2, K)}$
$t_0 = t[t == 0]$
$i \leftarrow 0$
while $i \leq 2K - 1$ **do**
    $I_\tau \leftarrow \text{zeros}(4, i)$
    $j \leftarrow 0$
    while $j \leq i$ **do** $I_\tau(:, j) \leftarrow e^{t(j)-t(i)} g_\epsilon(1:2, j)$
    end while
    $I_2(:, i) \leftarrow \text{trapz}(t(1:i), z(:, 1:i), 2)$
end while
$y \leftarrow x_c + [I_2; I_1]$

Remark 4.3.1. *Ultimately, we want to use this map to understand trajectories of the (3.3) system. However, the (3.13)-(3.17) system at $s = 0$ corresponds to the* final *time of the (3.3) system. Hence, we need to numerically extend the solution using ODE45 in order link the solutions back to the parameters (α, β).*

4.3.1 Error estimates

In Section 4.2, we justified that the error arising from the trapezoidal rule in the Γ calculation can be made arbitrarily small. One can quickly check that such a statement holds in this case with the same reasoning. However, we have to take into account the error due to the ODE45. In this case, the "error" in the W sense is much smaller than that induced by the approximation Λ^* on $[-R, R]$ (for instance) as we can set the absolute error tolerance to be small.

Lemma 4.3.2. *Let Λ^* denote our approximation to the true map Λ. Then*

$$\sup_{|t|<R} |\Lambda(z, w)(t) - \Lambda^*(z, w^*)(t)| \leq |\pi_s(z + \epsilon)| + 4R\epsilon + \delta$$

where $w \in W$, $\delta > 0$, $R > 0$ and $\|\cdot\|$ is the norm on W.

Proof. Consider the approximation, Λ^* supported on $t \in [-R, R]$, $R > 0$. Then define

$$I_s(w, \tau) = e^{A(t-\tau)} \pi_s(g_\epsilon(z^*(\tau) + w(\tau)) - g_\epsilon(z^*(\tau)) - \varphi(\tau))$$

and

$$I_c(w,\tau) = \pi_c(\varphi(\tau) + g_\epsilon(z^*(\tau)) - g_\epsilon(z^*(\tau) - w(\tau)))$$

Where $\mathcal{L}$ denotes $\Lambda(z,w) - \Lambda^*(z,w^*)(s)$ restricted to the stable and unstable parts respectively. To begin our analysis, suppose that $|t| \leq R$ and recall that $\Lambda^*(\tau)$ is supported on $[-R,R]$. Define $\mathcal{L}_{\text{stable}}$ as

$$\mathcal{L}_{\text{stable}}(t) = \int_{-\infty}^{t} I_s(w,\tau)d\tau - \int_{-R}^{t} I_s(w^*,\tau)d\tau = \int_{-\infty}^{-R} I_s(\tau)d\tau + \int_{-R}^{t} I_s(w,\tau) - I_s(w^*,\tau)d\tau.$$

Then $|I_s(w,\tau) - I_s(w^*,\tau)| = |e^{\tau-t}\pi_s(g(z^*(\tau)+w(\tau)) - g(z^*(\tau)+w^*(\tau))| \leq 2\epsilon e^{\tau-t}$ and thus

$$\left| \int_{-R}^{t} I_s(w,\tau) - I_s(w^*,\tau)d\tau \right| \leq 2\epsilon.$$

Moreover,

$$\begin{aligned}
\int_{-\infty}^{-R} I_s(\tau)d\tau &= \int_{-\infty}^{-R} e^{\tau+R}\pi_s(g_\epsilon(z+w(\tau)) - g_\epsilon(z) + Az + g_\epsilon(z))d\tau \\
&= \int_{-\infty}^{-R} e^{\tau+R}\pi_s(g_\epsilon(z+w(\tau)) + Az)d\tau \\
&\leq \int_{-\infty}^{-R} e^{\tau+R}\pi_s(\epsilon + Az)d\tau \\
&= \pi_s(\epsilon + Az)\int_{-\infty}^{-R} e^{\tau+R}d\tau \leq |\pi_s(\epsilon+Az)| = |(z_1+\epsilon, z_2+\epsilon, 0, 0)|
\end{aligned}$$

Similarly,

$$\mathcal{L}_{\text{center}}(t) = \int_{t}^{R} I_c(w,\tau) - I_c(w^*,\tau)d\tau + \int_{R}^{\infty} I_c(w,\tau)d\tau$$

and thus

$$|I_c(w,\tau) - I_c(w,\tau)| \leq 2\epsilon$$

implies that $\left| \int_t^R I_c(w,\tau) - I_c(w^*,\tau)d\tau \right| \leq 4\epsilon R$. On the other hand,

$$\int_{R}^{\infty} I_c(\tau) = \int_{R}^{\infty} \pi_c(g_\epsilon(\phi(\tau,z)) - g_\epsilon(\phi(t,z)+w(\tau)))d\tau.$$

As $w(t) \to 0$, exponentially quickly, it follows that

$$\left| \int_{R}^{\infty} I_c(\tau)d\tau \right| \leq \delta$$

for some $\delta > 0$.

□

We now consider the $|t| > R$ case.

Lemma 4.3.3. *If Λ^* is our approximation of the true map, Λ, then the following estimates hold.*

$$\sup_{t>R} e^{\frac{t}{2}} |\Lambda(z,w)(t) - \Lambda^*(z,w^*)(t)| \leq \sup_{t>R} e^{\frac{t}{2}} |\pi_s(z+\epsilon) + 2\epsilon(2 + e^{-2R} + e^{R-t}) + \delta|$$

and

$$\sup_{t<-R} e^{\frac{t}{2}} |\Lambda(z,w) - \Lambda^*(z,w^*)| \leq e^{-\frac{R}{2}} (3\pi_s(|\epsilon + z|) + 4\epsilon R + \delta)$$

for some $\delta, \delta' > 0$ arbitrarily small, $w \in W$, $\delta > 0$ and $\|.\|$ is the norm on W.

Proof. Suppose $t < -R$. Then

$$\mathcal{L}_{\text{stable}} = \left| \int_{-\infty}^{t} I_s(w,\tau)d\tau - \int_{-R}^{t} I_s(w^*,\tau)d\tau \right| = \left| \int_{-\infty}^{t} I_s(\tau)d\tau \right| \leq \pi_s(|\epsilon + z|)$$

and the center part satisfies

$$\begin{aligned}
\mathcal{L}_{\text{center}}(t) &= \left| \int_{t}^{\infty} I_c(w,\tau)d\tau - \int_{-R}^{R} I_c(w^*,\tau)d\tau \right| \\
&= \left| \int_{t}^{-R} I_c(w,\tau)d\tau + \int_{-R}^{R} I_c(w,\tau) - I_c(w^*,\tau)d\tau + \int_{R}^{\infty} I_c(w,\tau)d\tau \right| \\
&\leq |\pi_s(z-\epsilon)(-R-t) + 4\epsilon R + \delta| \leq \pi_s(z+\epsilon)| - R - t| + 4\epsilon R + \delta
\end{aligned}$$

Hence

$$\begin{aligned}
\sup_{t<-R} e^{\frac{t}{2}} |\mathcal{L}_{\text{stable}(t)} + \mathcal{L}_{\text{center}}(t)| &\leq e^{-\frac{R}{2}} (\pi_s(|\epsilon + z|) + 4\epsilon R + \delta) + 2\pi_s(|\epsilon + z|) e^{-\frac{R}{2}} \\
&\leq e^{-\frac{R}{2}} (3\pi_s(|\epsilon + z|) + 4\epsilon R + \delta)
\end{aligned}$$

On the other hand, if $t > R$, then

$$\begin{aligned}
\mathcal{L}_{\text{stable}}(t) &= \left| \int_{-\infty}^{-R} I_s(w,\tau)d\tau + \int_{-R}^{R} I_s(w,\tau) - I_s(w^*,\tau)d\tau + \int_{R}^{t} I_s(w,\tau)d\tau \right| \\
&\leq |\pi_s(z+\epsilon) + 2\epsilon(1 + e^{-2R}) + 2\epsilon(1 - e^{R-t})| \\
&\leq \pi_s(z+\epsilon) + 2\epsilon(1 + e^{-2R}) + 2\epsilon(1 + e^{R-t})
\end{aligned}$$

and

$$\mathcal{L}_{\text{center}}(t) = \left| \int_{t}^{\infty} I_c(w,\tau)d\tau \right| < \delta$$

hence,

$$\sup_{t>R} e^{\frac{t}{2}} |\mathcal{L}_{\text{stable}}(t) + \mathcal{L}_{\text{center}}(t)| \leq \sup_{t>R} e^{\frac{t}{2}} |\pi_s(z+\epsilon) + 2\epsilon(1+e^{-2R}) + 2\epsilon(1+e^{R-t}) + \delta|$$

implies that if z and ϵ are sufficiently small, then the error is not large.

□

The combination of Lemma's 4.3.2, 4.3.3 imply the following key proposition.

Proposition 4.3.4. *Suppose that $(X_1, X_2, Y_1, Y_2) : (-\infty, T) \to \mathbb{R}^4$ is a solution to (3.3.1) that satisfies the smoothness conditions (3.2.1). Let $z = (X_1, X_2, Y_1, Y_2)(T)$ and suppose that $\phi(t, z) \in B_\delta(0)$ for all $t \geq 0$ where $\phi(t, z)$ denotes the solution to the equations (3.13)-(3.16) that coincides with z at $t = 0$. Then the error in the map $h : \mathbb{R}^4 \to \mathcal{M}$ defined by $z \mapsto z + w(0)$ where w is the fixed point of $\Lambda^*(z, \cdot)(t)$, relative to $\Lambda(z, w)$ can be made arbitrarily small.*

Proof. Since $\phi(t, z) \in B_0(\delta)$, by [43, Theorem 7], it follows that $|\pi_s(z)|$ gets arbitrarily small. Then by Lemma's 4.3.2, 4.3.3, it follows that the error in the map defined by Λ^* can be made arbitrarily small by taking ϵ sufficiently small. □

The *key* point of Proposition 4.3.4 is that we can numerically establish an equivalence between trajectories that are eventually in some small ball about the origin and those on the center manifold. Moreover, by choosing our ϵ sufficiently small and our final time T sufficiently large, we can guarantee that the numerical approximation has arbitrarily small error.

Remark 4.3.5. *It is clear that the final iterate, w_N of our the approximation Λ^* will belong to W if we start with $w_0 \equiv 0$ by a simple induction argument.*

4.3.2 Numerical simulation

In this section, we focus on the practicalities of simulating Λ. First we note that if ϵ is *very* small, the algorithm may not converge to a fixed point. This is a simple consequence of g_ϵ being a cutoff function. I.e. if ϵ is small, then $g_\epsilon \equiv 0$.

Another natural question is what the initial choice, w_0 should be. Practically, we find that taking $w_0 \equiv 0$ leads to convergence in a reasonable number of iterations. Indeed, due to the global convergence property of this Picard-iteration based algorithm, any choice will eventually converge to a fixed point in a finite number of steps. Recall that a key hypothesis of (4.3.4) was that $|\pi_s(z)|$ was small. Hence if one has a solution to the equations (3.13)-(3.17) that is incomplete, the error in the Λ^* approximation may accumulate such that the iteration does not converge.

To make this more concrete, we give an example of iterating Λ^*, starting from a solution of the (3.3)-(3.5) system that satisfies the smoothness conditions (3.2.1).

Example 4.3.1. By (3.3.2), we can assume that a solution to the soliton equations (3.3)-(3.5) that satisfies (3.2.1) is characterised by a single parameter, $\gamma \in \mathbb{R}$. Let $k = 3$ and $\gamma \in [1.2, 1.5]$ and numerically integrate the system (3.3)-(3.5) up to time $T = 10$. One can then perform the change of variables for the system (3.13)-(3.17), recalling that the final

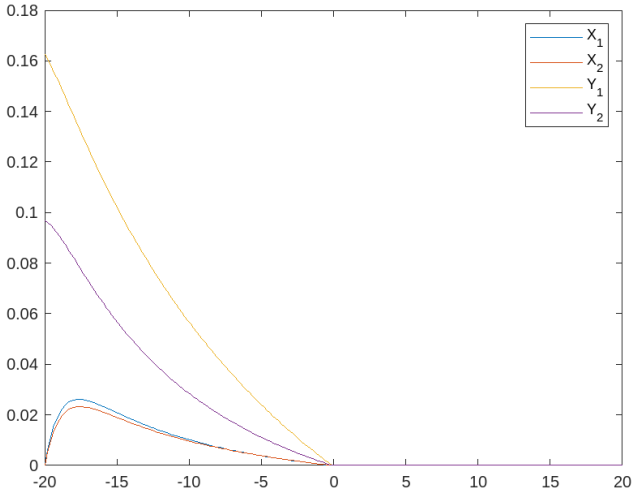


Figure 4.3: Result of 30-iterations of $\Lambda(\cdot)$ applied to $w_0 \equiv 0$, $\gamma = 1.5$.

time, T corresponds to $s = 0$ in these variables. Next, we take the initial guess, $w_0 \equiv 0$ for some $z \in \mathbb{R}^4$. For this example, we let $R = 20$, $K = 100$ and perform 30 iterations as in Algorithm 3. Figure 4.3 then shows the resulting solution curves for our final iterate, w_n.

Notice how Figure 4.3 shows that the w_n grows slowly as $s \to -\infty$ and is close to zero for $s \to \infty$, consistent with w_n belonging to the Banach space W defined in Section 4.3. Now consider the zoomed in region about 0 as in Figure 4.4.

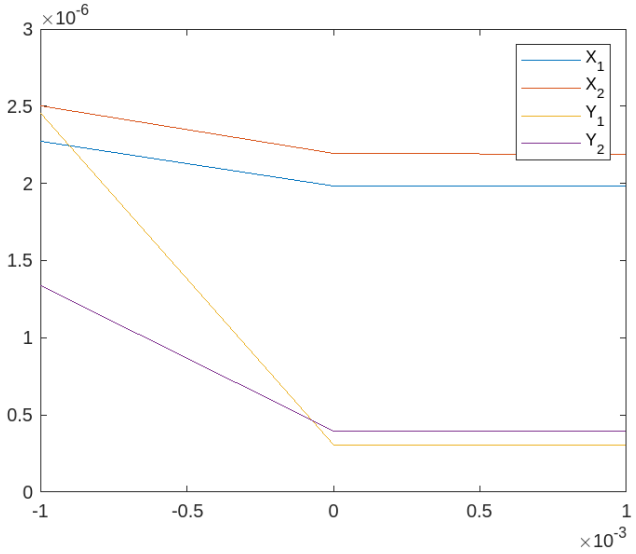


Figure 4.4: Result of 30-iterations of $\Lambda(\cdot)$ applied to $w_0 \equiv 0$, $\gamma = 1.5$ about $t = 0$.

Observe that $Y_1 \leq Y_2$ in Figure 4.4, consistent with this solution being complete. Furthermore, one can apply the same sequence of steps to $\gamma = 1.2$ to find that the corresponding center manifold trajectory satisfies $Y_1 > Y_2$, which is consistent with an incomplete solution by Proposition 4.1.3.

Chapter 5

A numerical study

In this section, we study solutions to the soliton equations (3.3)-(3.5) that satisfy the smoothness conditions, (3.2.1) numerically. We first establish the relationship between these solutions and the parameters (α, β). In particular,

Lemma 5.0.1. *The quantity $f_2(0)$ depends on both β in (3.3.1) and k as defined in (1.7). In particular, $f_2(0) = \sqrt{\frac{k}{2\beta}}$*

Proof. By Theorem 3.3.1, β is given by $\lim_{r\to 0} \frac{R_1}{r}$. Then,

$$\begin{aligned}\beta &= \lim_{r\to 0} \frac{R_1}{r} \\ &= \lim_{r\to 0} R_1' \\ &= \lim_{r\to 0} \frac{f_1'}{f_2^2} = \frac{k}{2f_2^2(0)}\end{aligned}$$

by L'Hôpital's rule and the boundary conditions (3.2.3). □

By considering γ, defined in (3.3.2), we fix $f_2(0) = 1$ and regard the solutions to the soliton equations (3.3)-(3.5) as functions of γ only, which is convenient for the purposes of numerical simulation. Indeed, from now on, we assume that $f_2(0) = 1$. Then, with the boundary conditions described in (3.2.3), we can numerically integrate the equations (3.3)-(3.5). To do so, we use the usual trick to simulate a system of second order ODEs by introducing the variables $(z_1, z_2, z_3, z_4, z_5, z_6) = (u_1, u_1', f_1, f_1', f_2, f_2')$, to obtain:

$$\begin{aligned} z_1' &= z_2, \quad & z_2' &= u'' \\ z_3' &= z_4, \quad & z_4' &= f_1'' \\ z_5' &= z_6, \quad & z_6 &= f_2''. \end{aligned}$$

We then use a Runge-Kutta method of order 4. A good description can be found in [9] though we include a brief discussion for completeness. Given an ODE $y' = F(r, y)$ with initial conditions, $y(0) = y_0$, the Runge-Kutta sequence of approximations is given by

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4),$$
$$t_{n+1} = t_n + h,$$

where

$$\begin{aligned} k_1 &= F(t_n, y_n), \\ k_2 &= F(t_n + \frac{1}{2}h, y_n + \frac{1}{2}hk_1), \\ k_3 &= F(t_n + \frac{1}{2}h, \frac{1}{2}hk_2), \\ k_4 &= F(t_n + h, hk_3), \end{aligned}$$

where the error in each step satisfies $\mathcal{O}(|h|^5)$ which gives an accumulated error of $\mathcal{O}(|h|^4)$. To get an explicit dependence on $u''(0)$, we use the standard trick of constructing a Taylor series approximation for $\epsilon > 0$ small. That is,

$$y(\epsilon) = y(0) + y'(0)\epsilon + \frac{y''(0)}{2}\epsilon^2$$

By [1, Theorem 6.1], if $f_1 > f_2$ somewhere on a trajectory, then the corresponding soliton is incomplete. On the other hand, to check if a solution is collapsed, we can take advantage of the equivalence established between solutions in U_0 and center manifold trajectories in Chapter 4 as follows.

1. Numerically integrate the soliton equations (3.3)-(3.5) corresponding to some (α, β), denoted $y = f(x)$ up to some time, $T > 0$ sufficiently large so that either $f_1 > f_2$ and the solution is incomplete (and thus no further steps are required), or the solution in the variables (3.13)-(3.16) lies in some small ball about the origin.

2. In the latter case, one can numerically calculate $\Lambda^*(\cdot)$ to determine the map $h : y(T) \mapsto \tilde{y}(T)$ as described in Section 4.3 where $\tilde{y}$ is the equivalent point on the center manifold.

3. We can then determine whether a solution is collapsed or otherwise by examining the quantity $K = h(Y_1) - h(Y_2)$. Indeed, if:

   - $h > 0$, then the solution is incomplete,
   - $h = 0$, then the solution is non-collapsed,
   - $h < 0$, then then the solution is collapsed and complete.

Remark 5.0.2. *Note that to derive rigorous bounds, one needs to be able to estimate the error in the numerical simulation precisely. For instance, the floating point and truncation errors. We do not make an attempt to do this here. Although, we note that this system is relatively well behaved so such estimates should be possible.*

For the purposes of exposition, let E_T be the error due to the numerical integration of the equations (3.3)-(3.5) at time $T > 0$. Then, the justification for steps (1-3) is as follows. If $f_1(T) > f_2(T) + E_T$, it follows that the solution is incomplete. Otherwise, by the properties of f_1, f_2 (described in Section 3.5), we can assume that the solution in the variables (3.13)-(3.17) for sufficently large $T > 0$ is in $B_\delta(0)$. By choosing ϵ, R and K as described in (4.3) appropriately, one can assume the error in the numerically approximated map, h is small and thus we can determine the behaviour of the solutions by applying Proposition 4.1.3.

5.0.1 Results

By numerically integrating the soliton equations (3.3)-(3.5) up to some time, $T > 0$ with respect to the boundary conditions (3.2.3) and γ, we observe three distinct behaviours, depending on the value of γ as shown in Figure 5.0.1.

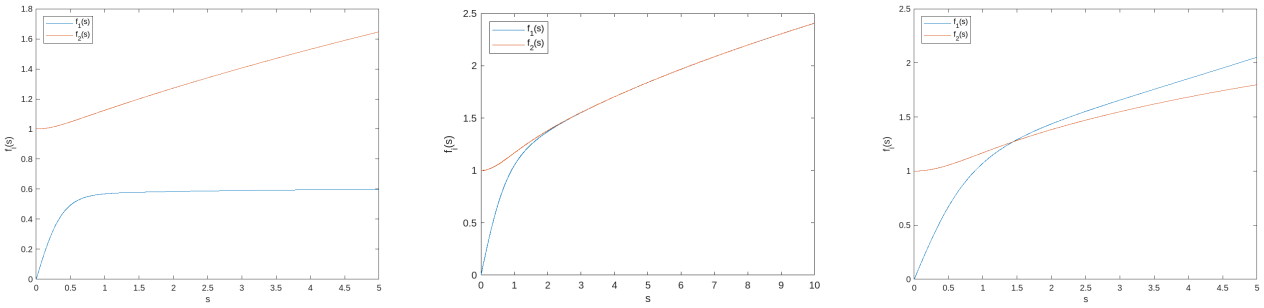


Figure 5.1: Collapsed, non-collapsed and incomplete solitons on $\mathbb{R}_{>0} \times SU(2)/\mathbb{Z}_3$.

Recall, that we can determine whether solutions are incomplete by examining $f_1 - f_2$. We find evidence that there exists critical bounds, (γ_0, γ_1) such that all solutions corresponding to $\gamma > \gamma_1$ are incomplete and $\gamma < \gamma_0$ are complete but collapsed. To do so, we use the technique described in steps (1-3), to find (γ_0, γ_1).

Conjecture 5.0.1. *By using the steps (1-3) we conjecture that the region U_0, lies in the following ranges for $\gamma \in \mathbb{R}$ where $k \in \mathbb{N}\backslash\{1, 2, 4\}$.*

k	$\gamma \in \mathbb{R}$ region containing U_0
3	$(1.2, 1.4)$
5	$(9, 10)$
6	$(14, 15)$

Table 5.1: Bounds of the set U_0 defined in Section 3

Remark 5.0.3. *The dependence of the soliton system (3.3) on the parameter, k, implies that it is possible to extend this argument to finite bounds for all $k \in \mathbb{N}\backslash\{1, 2, 4\}$.*

We now present evidence that U_0 is a single point and outline how one would go about proving this rigorously.

Suppose that one was able to estimate the numerical error precisely and is therefore able to prove (5.0.1) rigorously. For the purposes of exposition, let $\phi_\gamma(r)$ be a solution to the soliton equations (3.3)-(3.5) that satisfies the smoothness conditions (3.2.1) characterised by γ. Moreover, let $\phi_{\mathcal{L}(\gamma)}(s)$ be the corresponding solution in the variables (3.13)-(3.17). To show that there exists a unique γ such that the corresponding solution is non-collapsed, it suffices to show that the map,

$$\gamma \mapsto \lim_{s\to\infty} \phi_{\mathcal{L}(\gamma)}(s) \tag{5.1}$$

has a unique zero. Recall that complete solutions satisfy $\phi_{\mathcal{L}(\gamma)}(s) \in B_\delta(0)$ for $s \geq T$ where $T > 0$ is sufficiently large by the asymptotic properties in Section 3.5. Thus, we have an equivalence with trajectories on the center manifold $\mathcal{M}$. Therefore, let $\phi_{h(\mathcal{L}(\gamma))}(s)$ denote the equivalent solution on the center manifold. We know from our earlier analysis that trajectories on the center manifold satisfy,

$$\lim_{s\to\infty} \phi_{h(\mathcal{L}(\gamma))}(s) = \begin{cases} Y_2 = C, Y_1 = \infty & \text{if } Y_1 > Y_2, \\ Y_2 = C, Y_1 = 0 & \text{if } Y_1 < Y_2, \\ Y_2 = Y_1 = 0 & \text{if } Y_1 = Y_2 \end{cases}$$

by Proposition 4.1.3. Thus, the quantity $Q(T) := \frac{h(Y_1(T))}{h(Y_2(T))}$ *determines* the properties of the solution. Indeed, if $Q = 1$ then the solution is non-collapsed and if $0 \leq Q < 1$, it is collapsed. Hence, fix $T > 0$ such that we can apply the equivalence of center manifold trajectories and consider Q, Q' as functions of γ only. That is, $Q(\gamma) := Q(T) \geq 0$. To make this concrete, consider the following example.

Example 5.0.1. Let $k = 3$ for k corresponding to $\mathbb{Z}_k$ defined in (1.7). Then, we can simulate solutions to (3.3)-(3.5) up to time $T = 10$ for $\gamma \in [1.2, 1.3]$. Performing the change of variables to (3.13)-(3.17) and numerically calculating the map $h : \mathbb{R}^4 \to \mathcal{M}$, we can plot the values of each component individually with respect to γ as shown in Figure 5.2.

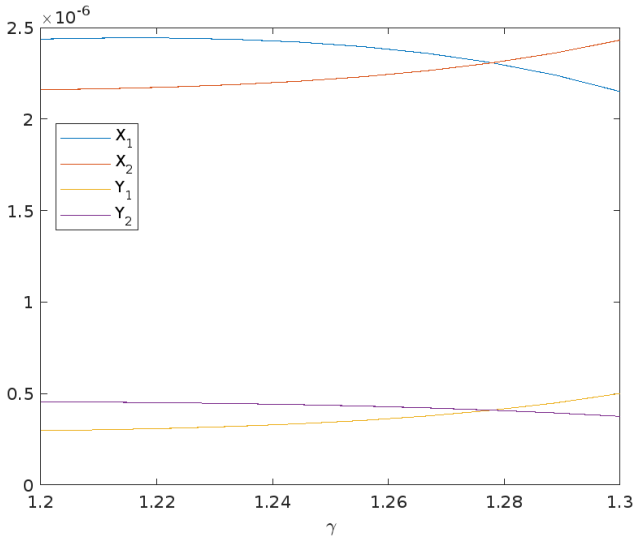


Figure 5.2: The values $(X_1, X_2, Y_1, Y_2)(T)$ on the center manifold at $T = 10$ for $\gamma \in [1.2, 1.3]$.

This figure indicates that there is a single point where $h(Y_1) = h(Y_2)$ and $h(X_1) = h(X_2)$ as anticipated. Figure 5.3 then illustrates the ratios $Q(\gamma) = \frac{h(Y_1)}{h(Y_2)}$, $Q'(\gamma) = \left(\frac{h(Y_1)}{h(Y_2)}\right)'$ of the corresponding trajectories on the center manifold.

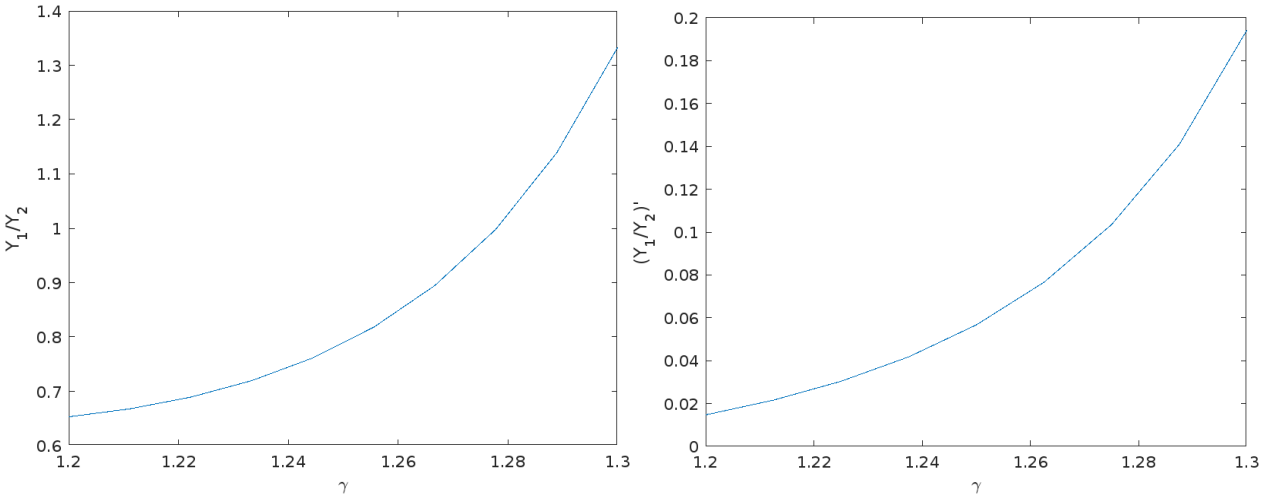


Figure 5.3: Ratios $Q(\gamma)$ and $Q'(\gamma)$ for $\gamma \in [1.2, 1.3]$.

Example 5.0.1 demonstrates that proving U_0 consists of a single point comes down to understanding the behaviour of Q and Q' as a function of γ in the case $k = 3$. Indeed, we claim one could use the proof sketch outlined below to prove that U_0 consists of a single point.

Proof. (sketch). Let $\gamma \in \tilde{U}$ as in (5.0.1). Then, we can fix a uniform time, T such that we can apply the equivalence of trajectories described in Section 4.3. By the error estimates in Section 4.3.1, we can suppose that the error in the map, $h : \mathbb{R}^4 \to \mathcal{M}$ is small. Hence we can consider the ratio $Q(\gamma) := \frac{h(Y_1(T))}{h(Y_2(T))}$ as a function of γ only. By numerically calculating $Q(\gamma)$ for $\gamma \in \tilde{U}$, we find that,

$$Q(\gamma) \in (0, C), \quad Q'(\gamma) \in (0, C')$$

for constants $C, C' > 0$. Then, by the mean value theorem, it follows that the function, $Q(\gamma)$ has a unique one. Thus, by Proposition 4.1.3, it follows that U_0 consists of a single point. □

This discussion shows that if one can control the numerical error precisely, then one would be able to prove that U_0 consists of a single point and therefore resolve Conjecture 1.1.1. Indeed, if the numerical errors in Example 5.0.1 could be made rigorous, this would show that U_0 consists of a single point for $k = 3$.

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